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Evaluation of a treatment of sleep-related problems in children with anxiety using a multiple baseline design

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Dissertation

**EVALUATION OF A TREATMENT OF SLEEP-RELATED PROBLEMS IN
CHILDREN WITH ANXIETY USING A MULTIPLE BASELINE DESIGN**

by

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Dedication

For the families that participated in this study, I learned so much from working with all of you. I was continually impressed by your engagement, determination, and humor and wish you all the best.

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ABSTRACT

Children with anxiety frequently experience sleep-related problems (SRPs) such as longer sleep onset latency, sleep dependence, and sleep anxiety. Despite the significant mental and physical health consequences of SRPs and the high prevalence of SRPs in school-aged children with anxiety, little research has examined sleep treatments for this population. This project utilized a single case multiple baseline design to evaluate the feasibility, acceptability, and preliminary efficacy of a four-session parent training intervention for reducing SRPs in youth with anxiety. Participants were ten children ($M=9.6$ years, range 8-12 years, 8 female) who met criteria for an anxiety disorder as well as a diagnosis of chronic insomnia. Throughout the study, parents tracked subjective child sleep onset latency (subjective SOL) using a daily sleep diary, children wore actigraphy monitors to capture daily objective sleep onset latency (objective SOL), parents completed a daily parent-child sleep interaction questionnaire, and parents and children completed qualitative interviews at post-treatment. Additional assessment of SRPs occurred at baseline, post-treatment, and one-month follow up. Results of the study showed that it was feasible to design and implement a four-session parent training intervention designed to treat SRPs in school-aged children with anxiety, and that this

intervention was acceptable to participants. Consistent with hypotheses, the majority of participants ($n=9$) no longer met criteria for chronic insomnia at post-treatment or follow-up ($n=6$), the frequency of SRPs was significantly lower at these time points, and the majority of participants ($n=7$) reported significant reductions in subjective SOL. However, only three participants showed significant reductions in objective SOL. Qualitative results helped to further clarify this discrepancy by demonstrating that participants experienced a shift in sleep perception across treatment (e.g., before treatment, normal objective SOL lengths were experienced as subjectively longer due to anxiety). Both quantitative and qualitative results demonstrated that parent behaviors (e.g., parental accommodation) and parent-child interactions (e.g., sleep dependence) both improved during the study. Future research should investigate changes in parent behaviors and parent-child interactions as potential mechanisms of treatment.

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List of Abbreviations

ADHD	Attention-Deficit Hyperactivity Disorder
ADIS-C/P	Anxiety Disorders Interview Schedule Child and Parent Report
CARD	Center for Anxiety and Related Disorders
CBT	Cognitive Behavioral Therapy
CRSP	Children's Report of Sleep Patterns Questionnaire
CSR	Clinical Severity Ratings
CSHQ	Children's Sleep Habits Questionnaire
DSM-IV	Diagnostic and Statistical Manual of Mental Disorders Fourth Edition
FACILIS	Family Accommodation Checklist and Interference Scale
FDA	Federal Drug Administration
GAD	Generalized Anxiety Disorder
ICSD	International Classification of Sleep Disorders
IRB	Institutional Review Board
M	Mean
MASC	Multidimensional Anxiety Scale for Children
OCD	Obsessive Compulsive Disorder
PDS	Pubertal Development Scale
PSIS	Parent-Child Sleep Interactions Scale
SAD	Separation Anxiety Disorder
SD	Standard Deviation
SE	Sleep Efficiency

SOL	Sleep Onset Latency
SRP	Sleep-Related Problem
TST	Total Sleep Time
WASO	Wake After Sleep Onset

EVALUATION OF A TREATMENT OF SLEEP RELATED PROBLEMS IN CHILDREN WITH ANXIETY USING A MULTIPLE BASELINE DESIGN

Introduction

The Importance of Sleep in Child Development and Functioning

Adequate quantity and quality of sleep is critical for healthy child development according to numerous medical organizations such as the American Academy of Sleep Medicine (Paruthi et al., 2016). Sleep is particularly important for school-aged children, generally defined as ages 6-12 years old, as this is a period of rapid physical development, emotional development, and brain maturation. Researchers have linked adequate sleep to a wealth of positive outcomes in school-aged children, including better emotion regulation, cognitive performance, alertness, and attention as well as greater positive affect (Busch, Altenburg, Harmsen, & Chinapaw, 2017; Paruthi et al., 2016). Additionally, a substantial body of research in the area of pediatric medicine links adequate sleep in children to better health outcomes such as healthier immune systems (Besedovsky, Lange, & Born, 2012) and better regulation of inflammatory pathways (Kim, Hakim, Kheirandish-Gozal, & Gozal, 2011). Although research indicates that most children and adolescents in the U.S. are sleeping less than recommended and would benefit from more sleep (Busch et al., 2017; Matricciani, Olds, & Petkov, 2012; Twenge, Krizan, & Hisler, 2017), researchers suggest that too much sleep can also negatively impact children's health and behavior (James & Hale, 2017).

Normative Sleep in Children

Sleep varies significantly across childhood development in terms of quantity, quality, and timing (Busch et al., 2017). For example, infants spend approximately 64% of their time asleep, but their sleep is distributed in inefficient short bursts. As children get older their sleep begins to consolidate and follow the light/dark cycles, with children sleeping about 14 hours a day at six months and decreasing to 11.4 hours a day at five years. Pediatric medical societies generally recommend that school-aged children get 10-11 hours of sleep each night (Paruthi et al., 2016; Vriend & Corkum, 2011). Although some research has found that school-aged children on average are sleeping approximately ten hours (Paruthi et al., 2016), one study found that school-aged children in a normative sample were only sleeping approximately eight hours a night using an objective measure of sleep (i.e., actigraphy; (Holley, Hill, & Stevenson, 2010). This study also found that it took children approximately 45 minutes to fall asleep (i.e., sleep onset latency) according to actigraphy data (Holley et al., 2010). In addition to quantity and quality of sleep, timing of sleep differs significantly across childhood and adolescence. For example, school-aged children generally go to bed and wake up earlier than adolescents (Knutson & Lauderdale, 2009; Owens, Spirito, McGuinn, & Nobile, 2000), and adolescents vary more widely in their wake times (e.g., sleep 1-2 hours later than children on the weekend; (Knutson & Lauderdale, 2009)). Although many children sleep well and navigate the developmental transition between sleep stages (e.g., from school-age to adolescence) effectively, some children develop sleep-related problems that can greatly impair daily functioning and development (Paruthi et al., 2016; Stein, Mendelsohn, Obermeyer,

Amromin, & Benca, 2001).

Sleep-Related Problems in Children

Sleep-related problems (SRPs), defined as non-developmentally appropriate sleep patterns, include a range of issues such as difficulty initiating sleep, difficulty sleeping alone, nighttime fears, and regular nighttime waking (Alfano, Ginsburg, & Kingery, 2007; Stein et al., 2001). SRPs can have wide-ranging detrimental effects on daily functioning in children in addition to far reaching mental and physical health consequences. Specifically, SRPs in childhood predict anxiety, depression, and attention difficulties in adolescence and adulthood, and are associated with other adverse consequences such as obesity, impulsivity, accidental injuries, and overall impaired health (Beebe, 2011; Chaput et al., 2016; Shanahan, Copeland, Angold, Bondy, & Costello, 2014; Smaldone, Honig, & Byrne, 2009; Vriend et al., 2013). SRPs have also been found to significantly disrupt family functioning and home life (e.g., relationships with siblings (Alfano et al., 2007)). SRPs are quite common in childhood, with approximately 20% of a non-clinical sample of four- to twelve-year-old children experiencing a SRP at least once a week (Stein et al., 2001).

Sleep-Related Problems in Children with Anxiety

Sleep-related problems are particularly co-prevalent in youth who are also experiencing anxiety disorders; in fact, numerous studies indicate that approximately 85% of children with anxiety demonstrate significant sleep problems (Alfano et al., 2007; Alfano, Pina, Zerr, & Villalta, 2010; Chase & Pincus, 2011; Ivanenko & Johnson, 2008). However, much of the sleep disturbance in children with anxiety appears to be related to

SRPs such as sleep anxiety, bedtime resistance, or reduced sleep quality rather than significant differences in objective sleep metrics (Alfano, Patriquin, & De Los Reyes, 2015). Indeed, children with anxiety appear to have patterns of sleep similar to typically developing youth with the exception of slightly longer sleep onset latencies (i.e., time to fall asleep) and increased sleep anxiety (Alfano et al., 2015; Alfano, 2018; Cousins et al., 2011; Fletcher et al., 2018; Forbes et al., 2008). One study also found that school-aged children with anxiety were more likely to demonstrate bedtime resistance and require parental assistance at night and to wake in the morning (Price, Farrell, Donovan, & Waters, 2019).

SRPs are highly comorbid with separation anxiety disorder (SAD) and generalized anxiety disorder in particular (GAD; (Alfano et al., 2007; Fletcher et al., 2018)). It has been hypothesized that sleep problems in children with SAD are most linked to difficulties separating from parents at bedtime and difficulties self-soothing in children with GAD. One study found that school-aged children with GAD have higher levels of SRPs according to child and parent report, but no significant differences in objective sleep measures assessed with actigraphy compared to the healthy control group (Alfano et al., 2015). The relationship between anxiety and SRPs appears to be stronger in children than adolescents (Donovan, Spence, & March, 2017). It is possible that this difference is due to a higher prevalence of separation fears and sleep-related fears (e.g., fear of the dark) in younger children. SRPs are associated with higher levels of anxiety, although it is unclear if SRPs exacerbate anxiety, if anxiety causes SRPs, or if the relationship is bidirectional (Alfano et al., 2007; Busch et al., 2017). It is also likely that

the high comorbidity between SRPs and anxiety can be attributed to shared risk factors and mechanisms (Busch et al., 2017), which may also suggest that similar treatments may be helpful in treating both sleep and anxiety problems.

Treatment of Sleep-Related Problems in Children

There is a strong body of evidence indicating that SRPs can be successfully treated in children using a range of behavioral techniques (Bourchtein, Langberg, & Eadeh, 2019; Johnson & Mindell, 2011; Mindell, Kuhn, Lewin, Meltzer, & Sadeh, 2006; Paine & Gradisar, 2011; Tikotzky & Sadeh, 2010). Such treatments generally include elements such as providing children and parents with psychoeducation about sleep as well as sleep hygiene techniques. More specifically, research typically supports that clinicians implement the “ABCs of SLEEPING” as outlined by Stephanie Allen and colleagues (2016): 1) age appropriate bedtimes and wake-times with consistency, 2) schedules and routines, 3) location, 4) exercise and diet, 5) no electronics in the bedroom or before bed, 6) positivity 7) independence when falling asleep, and 8) needs of child met during the day (Allen, Howlett, Coulombe, & Corkum, 2016). Research suggests that sleep treatments involving these types of elements are effective in treating SRPs in a variety of age ranges (Bourchtein et al., 2019; Johnson & Mindell, 2011; Mindell et al., 2006; Paine & Gradisar, 2011; Tikotzky & Sadeh, 2010).

Although these studies have shown promising results, there is still a need for well-designed randomized control trials to determine the best methods of treatment for school-aged children with SRPs (Bourchtein et al., 2019; Busch et al., 2017; Meltzer, 2017). This is critical because school-aged children (ages six to twelve) have very different age

appropriate sleep recommendations and needs than infants, young children, or adolescents. Given that school-aged children need 10-11 hours of sleep each night but do not always receive it (Holley et al., 2010; Paruthi et al., 2016; Vriend & Corkum, 2011), treatments need to be tailored to helping these youth increase their sleep quality and quantity. Understanding of sleep needs in this age group is also important because the specific skills taught to school-aged youth may differ from other age groups. For example, as adolescents often sleep late on the weekends and wake very early during the week, it may be especially important to focus on consistent wake times in adolescents with sleep difficulties (Knutson & Lauderdale, 2009; Owens et al., 2000). On the other hand, school-aged children may have more consistent wake times naturally, but may need more support in developing sleep independence (e.g., falling asleep without parents present).

Treatment of Sleep-Related Problems in Children with Anxiety

Research on sleep treatments for school-aged children with anxiety is even more scarce (Bourchtein et al., 2019; Meltzer, 2017). As children with anxiety often have different SRPs than school-aged children without anxiety (Alfano et al., 2010; Alfano et al., 2015; Leahy & Gradisar, 2012), current treatments may not effectively address sleep in this population. For example, developing age-appropriate sleep independence may be particularly difficult for children with separation anxiety and may require different skills than treatment of SRPs in children without anxiety. SRPs in children with anxiety may also be initiated and maintained by different factors than SRPs in other populations (Peterman, Carper, & Kendall, 2015). For example, children with anxiety may be more

likely to have SRPs related to parental attention to and accommodation of non-adaptive sleep and avoidance behaviors (Peterman et al., 2015), whereas children with attention-deficit hyperactivity disorder (ADHD), may be more likely to have movement sleep disorders related to an imbalance of dopamine (Hening, Allen, Earley, Picchietti, & Silber, 2004). In children with anxiety, parental accommodations (e.g., allowing children to sleep with the light on) may reinforce children's anxiety (e.g., fear of the dark), making it even more difficult for the child to transition back to adaptive sleep habits. For this reason, it is unclear whether existing treatments effectively target diagnosis-specific mechanisms in children with anxiety and SRPs.

The research that has been conducted on treatment of SRPS in youth with anxiety thus far has indicated that cognitive behavioral therapy for anxiety may be effective at improving SRPS including bedtime resistance and sleep anxiety (Peterman et al., 2016). However, in this study, youth continued to have residual sleep problems even after the treatment for anxiety ended, further highlighting the need for the development of more intensive treatments that are specifically tailored to youth with comorbid sleep and anxiety problems. Furthermore, treatments for youth with anxiety and sleep problems could be enhanced if there were treatment components that directly targeted improving sleep hygiene, reducing parental accommodation, and improving parental reinforcement of adaptive sleep behaviors.

Current Study

The overarching purpose of the present study was to substantively contribute to the existing literature on the effective treatment of youth with anxiety and comorbid sleep

problems by designing and testing a four-session parent training intervention for reducing sleep-related problems (SRPs) in this population. The present study utilized a multiple baseline design and incorporated quantitative assessment of changes in youth's sleep and anxiety symptoms as well as exploratory qualitative evaluation of parent and child perspectives of treatment. The study had five specific aims that collectively evaluated the feasibility and acceptability of designing and implementing a brief parent training intervention, changes in parent-child sleep interactions (e.g., sleep dependence) during treatment, and the effect of treatment on sleep (e.g., sleep onset latency) and non-sleep (e.g., anxiety) outcomes.

The first aim was to test the feasibility of designing and implementing a four-session parent training intervention designed to treat SRPs in school-aged children with anxiety. It was hypothesized that all participants would complete treatment within approximately four weeks as intended, and that treatment fidelity would be high according to two independent evaluators.

The second aim was to determine the effects of the intervention on parent-child sleep interactions and parenting behaviors. It was hypothesized that, according to single case data, the majority of parent participants would report significant large increases in sleep reinforcement (e.g., rewarding adaptive sleep behaviors) and significant large decreases in sleep conflict (e.g., arguing about bedtime) and sleep dependence (e.g., co-sleeping) during treatment. It was also hypothesized that, according to questionnaire data, overall rates of parental accommodation (i.e., total FACLIS score) and rates of

ineffective discipline (i.e., total Parenting Scale score) would be significantly lower at post-treatment and follow-up as compared to baseline.

The third aim was to evaluate the effects of the intervention on sleep outcomes using questionnaire data collected at baseline, post-treatment, and follow-up assessments and single case data collected daily throughout baseline, treatment, post-treatment, and follow-up weeks. It was hypothesized that the majority of participants would no longer meet criteria for chronic insomnia at post-treatment or follow-up. It was also hypothesized that according to questionnaire data, the frequency of SRPs such as sleep anxiety and bedtime resistance would be significantly lower at post-treatment and follow-up than baseline. Next, it was hypothesized that, according to single case data, the majority of participants would demonstrate significant large decreases in self-reported (i.e., sleep diary) sleep onset latency (i.e., subjective SOL) and actigraphy-measured sleep onset latency (i.e., objective SOL) during treatment.

The fourth aim was to evaluate the effects of the intervention on general functioning and non-sleep symptoms using questionnaire data collected at baseline, post-treatment, and follow-up as well as reward learning task data collected at baseline and post-treatment. It was hypothesized that, overall anxiety (i.e., total MASC score) would be significantly reduced at post-treatment and follow-up as compared to baseline. It was also hypothesized that, according to questionnaire data, overall functioning (as measured by the Pediatric Symptom Checklist and the Brief Impairment Scale) would be significantly improved at post-treatment and follow-up as compared to baseline. Finally, it was hypothesized that, according to data from the reward learning task, reward

sensitivity would be high in this sample and would not significantly differ from baseline to post-treatment.

The fifth aim was to use qualitative methods to describe and further explore parent and child perspectives on the process of engaging in treatment and changes in sleep and non-sleep symptoms during treatment. It was hypothesized that parents and children would describe the treatment as acceptable and describe changes in parent-child sleep interactions and parenting behaviors during treatment (e.g., reductions in sleep dependence). It was also hypothesized that parents and children would describe improvements in sleep outcomes (e.g., sleep onset latency) and non-sleep outcomes (e.g., anxiety) during treatment.

Treatment Development

Rationale for Current Treatment Development. Developing an effective intervention for SRPs in children with anxiety is critical given the high prevalence SRPs in this population, the significant consequences of SRPs, and the fact that SRPs in children with anxiety appear to differ from those in other populations (Alfano et al., 2015). Additionally, for children with SRPs and anxiety, factors related to anxiety (e.g., avoidance, parental accommodation) may play an important role in maintaining SRPs. In order to address all of these concerns, a brief parent training treatment was developed that included sleep treatment elements (e.g., sleep hygiene), but also components of anxiety treatments (e.g., rewards, exposure). Parent training was selected as the treatment modality of choice because parents are still highly involved in bedtimes of school-aged children and their own behaviors play a significant role in their child's sleep (Palmer,

Clementi, Meers, & Alfano, 2018). For example, parental accommodations such as co-sleeping may reinforce avoidance and maintain SRPs by inadvertently teaching the child that they cannot tolerate sleeping alone. On the other hand, parents can promote healthy sleeping habits by helping children engage in good sleep hygiene (e.g., waking children up at the same time every day) and making sleep a more positive experience (Allen et al., 2016) by reinforcing adaptive child behaviors (e.g., providing rewards for steps toward independent sleeping).

The intervention was designed to be brief (four sessions) and intensive (90-minute weekly sessions) in order to achieve healthy functioning quickly and so that the treatment could potentially be implemented in a range of settings (e.g., while on the waitlist for outpatient anxiety treatment). Previous studies have shown that short term (three- to five-session) behavioral treatments for sleep result in improvements shortly after treatment initiation in a variety of contexts (Mindell, Telofski, Wiegand, & Kurtz, 2009; Montgomery, Stores, & Wiggs, 2004; Palermo, Beals-Erickson, Bromberg, Law, & Chen, 2017; Papadopoulos et al., 2019). For example, one study demonstrated that four two-hour sessions focused on psychoeducation, sleep hygiene, and reinforcement of adaptive sleep strategies were found to improve subjective sleep duration, sleep quality, sleep onset latency, stress, and anxiety in adolescents (Paavonen, Huurre, Tilli, Kiviruusu, & Partonen, 2016). Additionally, a five-session phone- and workbook-based parent training intervention demonstrated efficacy in decreasing SRPs in school-aged children with insomnia with and without comorbid ADHD (Corkum et al., 2016). However, to our knowledge, there has yet to be a study of a brief parent training

intervention for school-aged children with SRPs and anxiety, and there is currently a significant lack of research on the most effective ways of directly treating SRPs in children with anxiety (Bourchtein et al., 2019; Caporino et al., 2015).

Process of Current Treatment Development. Before designing the intervention for this study, literature on normative sleep in school-aged children as well as SRPs in school-aged children with anxiety was reviewed in order to determine appropriate treatment targets. Specifically, research suggested that treatment should aim for reducing sleep onset latency, sleep anxiety, and sleep dependence, all SRPs particularly prevalent and interfering in children with anxiety (Alfano et al., 2015; Alfano, 2018; Cousins et al., 2011; Fletcher et al., 2018; Forbes et al., 2008). Next, the evidence-based sleep treatment literature for children was reviewed and effective treatment components were identified (e.g., Allen et al., 2016) and included in treatment if applicable to the SRPs identified to be prevalent in school-aged children with anxiety. For example, sleep hygiene components such as regular bedtime routines and working towards independent sleep were determined to be relevant to the SRPs of longer sleep onset latency and sleep dependence common in children with anxiety and were included in the treatment. Treatment components that were less applicable to this population were often included in sleep psychoeducation, but not as individual treatment components. For example, as school-aged children are less likely to have the inconsistent wake times common to adolescents (Knutson & Lauderdale, 2009), setting consistent wake times was addressed briefly in psychoeducation materials but was not included as a separate treatment component.

In order to address factors related to anxiety (e.g., avoidance, parental accommodation) in children with comorbid SRPs, the evidence-based anxiety treatment literature was also reviewed and applicable skills included in treatment. Parent training interventions designed to address childhood anxiety often include treatment components related to anxiety psychoeducation, changing parent behaviors (e.g., reducing parental accommodation), reducing child avoidance (e.g., exposure), and reinforcing adaptive behaviors (e.g., rewards; (Merson & Chu, 2011)). All of these elements were determined to be applicable to children with comorbid anxiety and SRPs and were included in the current treatment. For example, psychoeducation was necessary to help parents understand the connection between avoidance and anxiety (e.g., avoiding sleeping alone makes independent sleeping more frightening). Additionally, as SRPs for children with anxiety (e.g., sleep dependence) often involve parent behaviors, skills to help parents reduce accommodation and help their child face their fears were applicable to this population. Similarly, rewards were included in the current treatment because they are commonly used in child anxiety treatment to help children overcome their fears (Kendall et al., 2005) and may help to shift bedtime to a more positive experience, an important part of sleep treatment (Allen et al., 2016).

After all of the applicable treatment components were identified, four 90-minute sessions were designed so that each new skill taught could build progressively upon prior skills. For example, sleep psychoeducation was introduced before sleep hygiene in order to help parents understand the importance of shifting sleep habits. Similarly, learning to reinforce adaptive sleep behaviors (e.g., use rewards) was placed before reducing

avoidance (e.g., exposure) so that parents could use rewards to reinforce when children faced their fears. The resulting parent training program targeting SRPs in children with anxiety is described in detail below (see methods section).

Methods

Participants

Participants included children with anxiety and at least one parent recruited during standard clinic flow of treatment-seeking families at the Boston University Center for Anxiety and Related Disorders (CARD). Interested families were required to meet the following inclusion criteria to be eligible for enrollment: 1) child was between the ages of eight and twelve years old, 2) child met Diagnostic and Statistical Manual of Mental Disorders 5th edition (DSM-5) criteria for a primary anxiety disorder (American Psychiatric Association, 2013), 3) child met the International Classification of Sleep Disorders 3rd Edition (ICSD) criteria for chronic insomnia (American Academy of Sleep Medicine, 2014), 4) at least one parent was present at bedtime and able to attend parent training sessions, 5) family had not engaged in sleep treatment for the child in the past year, and 6) child was not taking medication or engaged in other treatment for sleep or anxiety for the duration of the study. As depression, sleep apnea, and attention-deficit hyperactivity disorder (ADHD) are associated with other types of sleep disorders (e.g., parasomnias rather than chronic insomnia (Meltzer, 2017)), children were screened and excluded from the study if they met criteria for any of these disorders.

Seventeen families were screened for the study. Six families were ineligible

because they: did not respond to outreach efforts ($n=2$), the child declined participation ($n=1$), the child did not meet screening inclusion criteria for chronic insomnia ($n=1$), the child met screening exclusion criteria due to sleep apnea ($n=1$), or the child was currently involved in anxiety treatment ($n=1$). One family met initial screening criteria and consented to the study, but it was determined during the baseline assessment that the child met criteria for ADHD, so this family was not enrolled. The remaining ten families were enrolled in and completed the study. One participant completed baseline, treatment, and post-treatment, but was lost to follow-up; the others completed all study components.

Child participants consisted of ten children aged 7.96 to 12.39 years old ($M=9.64$, $SD=1.47$) who had not yet reached puberty (Pubertal Development Scale Score $M=1.57$, $SD=.43$). The majority of the participants were female ($n=8$), non-Hispanic ($n=7$), and White ($n=9$). All of the participants met criteria for chronic insomnia and a primary anxiety disorder (i.e., generalized anxiety disorder ($n=9$) or social anxiety disorder ($n=1$)). Four participants were also assigned other co-morbid diagnoses (i.e., social anxiety disorder ($n=3$), oppositional defiant disorder ($n=1$)). The existence of a behavioral sleep disorder (i.e., chronic insomnia) without comorbid parasomnias or sleep apnea was confirmed during clinical interviews. The majority of participants ($n=7$) had been experiencing sleep difficulties for at least four years, and the other participants ($n=3$) reported sleep difficulties for nine months to three years. At baseline, participants generally met criteria by exhibiting difficulties with sleep initiation ($n=8$), sleep dependence ($n=6$), bedtime resistance ($n=2$), and/or sleep maintenance ($n=2$) for an average of 5.45 days a week, all above the clinical cutoff of three days (Table 1). At

baseline, participants demonstrated the following consequences of poor sleep: irritability/sadness ($n=9$), impaired attention/more errors/hyperactivity ($n=8$), sleep anxiety ($n=7$), impaired social/family/academic functioning ($n=7$), daytime sleepiness ($n=7$), and/or fatigue ($n=3$). Please refer to Table 1 for all diagnostic criteria.

Each child participant had a parent who attended all sessions, completed the phone check-ins, and completed all assessment components; analyses were completed with data from these parents. Most of the parents in this study were female ($n=9$) and highly educated (all completed a college degree or higher). Parents were 38.63 to 52.32 years old ($M=43.13$, $SD=4.16$). The majority of parents were married ($n=9$) and had a household income of over \$100,000 ($n=8$).

Study Design

The current study employed an explanatory concurrent mixed methods design (Creswell, Klassen, Plano Clark, & Smith, 2011) in which both quantitative data and qualitative data were collected to address the research aims (e.g., evaluate the effect of treatment on sleep outcomes). The quantitative component of the study consisted of a single case non-concurrent multiple baseline design across subjects where participants were randomized to baseline length (i.e., multiple baseline) as they enrolled in the study (i.e., non-concurrent). Participants were randomized using a random numbers table by a graduate-level clinician not involved the study. Each participant was assigned to complete either a one-week ($n=3$), two-week ($n=3$), or three-week ($n=3$) baseline. Participant ten (P10) was non-randomly assigned to a one-week baseline due to time constraints. After completing the baseline, parents completed four sessions of treatment

over approximately four weeks. Objective sleep (i.e., child actigraphy), subjective sleep (i.e., child sleep diary), and parent-child sleep interactions (e.g., sleep dependence) were tracked daily during baseline and treatment weeks as well as for one week at post-treatment and one week at follow-up assessment (Figure 1).

In addition to daily data collected as part of the single case design, questionnaires on anxiety, sleep, and other related topics were completed by parents and children at baseline, post-treatment, and follow-up assessments (Figure 1). At the post-treatment assessment, all child participants and their parents also participated in a semi-structured interview on treatment process (e.g., skills learned) and outcomes (e.g., changes in sleep). Interviews were transcribed and through the use of qualitative methods key themes were identified to provide in-depth insight into the context and meaning of participants' experiences (e.g., changes in sleep dependence) and possible explanations for quantitative findings (Creswell et al., 2011). Please refer to Figure 1 for timing of study components.

Single Case Design. A single case non-concurrent multiple baseline design across subjects was selected for this study because with this design each participant's baseline acted as their own control condition, meaning that outcomes of treatment could be determined with a much smaller sample size (Barlow, Nock, & Hersen, 2009; Rhoda, Murray, Andridge, Pennell, & Hade, 2011). It also potentially allows for examination of mechanisms of treatment, as individual behaviors (e.g., sleep reinforcement) and outcomes (e.g., sleep onset latency) can be tracked to determine whether changes in outcomes occur after and only after behaviors change. Additionally, a multiple baseline

design was employed because this type of design is recommended when it may not be ethical or possible to withdraw treatment completely or when interventions may have lasting effects (Kazdin, 2011). In the current study, the intervention was parent training; once parents are trained in specific skills, this knowledge cannot be removed, and it was hypothesized that the intervention would have lasting effects on parent behavior.

Single case designs are considered scientifically rigorous if they meet specific evidence standards (Kratochwill et al., 2010). The current study met evidence standards for a single case design because: 1) the independent variable (i.e., treatment) was manipulated to examine effects on dependent variables, 2) each outcome variable (e.g., chronic insomnia diagnosis) was measured systematically over time by more than one assessor and inter-rater reliability was acceptable, 3) the study attempted three times to demonstrate an intervention effect (i.e., three participants in each baseline condition), and 4) each phase (i.e., baseline, treatment) for each participant had at least five data points. A sample size of 9-10 is generally considered appropriate for a multiple baseline with three different baselines (Barlow et al., 2009) and is significantly higher than the modal number of three to four participants in most multiple baseline studies (Shadish & Sullivan, 2011).

Study Procedures

This study consisted of a baseline assessment, a variable-length baseline (one, two, or three weeks), four sessions of parent training, a post-treatment assessment (one week after treatment completion), and a follow-up assessment (one month after treatment completion). Please see Figure 1 for an overview of assessment time points and

measures. All study procedures were approved by the Boston University Institutional Review Board (IRB).

Recruitment. Eligible families were recruited from the Boston University Center for Anxiety and Related Disorders (CARD), an outpatient anxiety specialty clinic. When families called CARD as part of general clinic practices, the phone screener determined whether families were calling regarding a child between the ages of eight and twelve years old and briefly asked about anxiety and sleep concerns. If families endorsed having a child with anxiety and sleep concerns and were potentially interested in participating in the study, the primary investigator (Lydia Chevalier) called the family to complete a more in-depth phone screen. The in-depth phone screen included questions on symptoms of chronic insomnia (e.g., difficulties with sleep initiation), frequency and duration of chronic insomnia symptoms (i.e., three times a week for at least three months), anxiety symptoms (e.g., likely to meet criteria for an anxiety disorder), and questions about our study exclusion criteria (e.g., likely to meet criteria for sleep apnea). If families met criteria for inclusion in the study according to this initial screening (e.g., likely to meet criteria for chronic insomnia and anxiety diagnoses, and unlikely to meet exclusion criteria), they were asked to come in to CARD to consent to the study and complete a baseline assessment.

Baseline Assessment. During the baseline assessment conducted at CARD, study eligibility was confirmed by the primary investigator (Lydia Chevalier) through the administration of the Anxiety Disorders Interview Schedule Child and Parent (ADIS-IV-C/P) version (Silverman & Albano, 1996) to determine whether the child met criteria for

a primary anxiety disorder, and through the administration of the International Classification of Sleep Disorders (ICSD) Clinical Interview to determine whether the child met ICSD criteria for a diagnosis of chronic insomnia (American Academy of Sleep Medicine, 2014). Additionally, parents and children completed questionnaires on sleep, anxiety, and other related topics through REDCap (Research Electronic Data Capture), a secure, web-based software platform designed to support data capture for research studies (Harris et al., 2009). Children also completed a reward learning task to determine reward learning sensitivity. After study eligibility was confirmed, each participant was randomly assigned to begin tracking symptoms and start treatment in one, two, or three weeks. Participant ten was non-randomly assigned to a one-week baseline due to time constraints. During the baseline weeks children wore actigraphy monitors to measure objective daily sleep outcomes and parents completed a daily child sleep diary and a daily Parent-Child Sleep Interactions Scale (Alfano, Smith, Reynolds, Reddy, & Dougherty, 2013).

Treatment. After completing the one, two, or three weeks of baseline, each parent completed four individual 90-minute parent training sessions led by the primary investigator (Lydia Chevalier) at CARD within approximately four weeks. After each session, parents were instructed to change one to two behaviors personalized to their child's needs as homework (e.g., initiate a reward system). Four weekly 30-minute parent phone check-ins with the primary investigator (Lydia Chevalier) were also provided to address any difficulties in implementing strategies. Phone check-ins were generally completed several days after the weekly session so that parents would have time to try the

skills learned during the session and bring any questions or problems to the check-in call. During treatment weeks, children wore actigraphy monitors to measure objective daily sleep outcomes and parents completed a daily child sleep diary and a daily Parent-Child Sleep Interactions Scale (Alfano et al., 2013). Each of the four treatment sessions are described in detail below and quotes that illustrate treatment components can be seen in Table 2.

Treatment Session Content

Session 1: Psychoeducation and Sleep Hygiene. During the first session, parents were provided with psychoeducation on normative sleep in children and the importance of sleep in child development. Specifically, parents were provided with a handout (“*Sleep in School-Aged Children (6-12 Years)*”) developed by Mindell and Owens (2003) that informed them that school-aged children require 10-11 hours of sleep per night and that poor sleep impacts mood, behavior, and cognitive ability. This information was presented at the beginning of treatment as many parents are unaware of best sleep practices for each age range (McDowall, Galland, Campbell, & Elder, 2017), and because parental buy-in on the importance of sleep is necessary for skill implementation later in treatment. Also included in the handout by Mindell and Owens (2003) and in handouts developed by the University of Kansas (“*Taking Charge of Your Child’s Sleep*”) and Children’s Hospital of Orange County (“*Sleep Hygiene for Children*”), was sleep hygiene information (e.g., regular bedtime routine and sleep schedule, no screens before bed). Sleep hygiene components were included in treatment so that parents had concrete changes that they could make to improve their child’s sleep. Sleep hygiene was tailored for this population

with a greater focus on changes (e.g., regular relaxing bedtime routine) that would help reduce their most common SRPs (e.g., sleep dependence, longer sleep onset latency) rather than things that are less commonly a concern for this population (e.g., regular wake times).

Anxiety Psychoeducation. During the first session, parents were also provided with psychoeducation on anxiety and the role of parents in addressing child anxiety. For example, parents were provided with handouts developed for an anxiety parenting group conducted at an anxiety specialty outpatient treatment program (Merson & Chu, 2011). These handouts focused on psychoeducation on anxiety (“*Anxiety is Like a False Alarm*”), the role of parents in maintaining and addressing anxiety (“*Where Do Parents Fit In?*”), and introduced parents to the concept of “parenting traps” (“*Types of Parental Responses*”, e.g., accidentally reinforcing anxiety by agreeing with it). Anxiety psychoeducation was presented at the beginning of treatment because understanding the role of anxiety in sleep difficulties was necessary before directly addressing sleep difficulties. For example, two children (one with anxiety and one without) may have the same SRP (e.g., bedtime resistance) but the child with anxiety may be resisting for different reasons (e.g., separation anxiety at bedtime) than the child without anxiety (e.g., jealousy that older siblings gets to stay up later). Understanding the cause of the SRP is important to most effectively addressing it. In the former case, the child may need to slowly work through exposures to reduce anxiety related to separation at bedtime before this SRP is addressed, while the child without anxiety may be able to more quickly reduce bedtime resistance with the use of rewards. This section of treatment was tailored

for this population by using examples of anxiety related to sleep (e.g., anxiety about separating from parents at bedtime).

Session 2: Reinforce Adaptive Sleep Behaviors. During the second session, parents were provided with psychoeducation and skills for reinforcing adaptive child behaviors. This treatment component included handouts such as “*Empathize and Encourage*”, “*Tools for Effective Praise*”, and “*Defining Successful Rewards*” (Merson & Chu, 2011). These skills were included in treatment because parent behaviors can be used to reinforce adaptive sleep behaviors. The concept of reinforcement of desirable and brave behaviors is regularly taught in anxiety treatments, where parents are instructed to give positive attention to adaptive behaviors and ignore maladaptive behaviors so as not to accidentally reinforce them (Kendall et al., 2005). Using behavioral reinforcement such as rewards or praise may help to shift bedtime to a more positive experience (Allen et al., 2016), in addition to providing extra motivation to children. This extra motivation is especially important for children with anxiety as they may have a more difficult time changing their behaviors due to sleep-related fears and may not see their behaviors (e.g., co-sleeping) as problematic and in need of change (Alfano, 2018). All of the skills in this session were tailored to this population by making examples and homework assignments specific to sleep anxiety. For example, parents were instructed to use empathize and encourage statements such as “I know that you are worried about not being able to fall asleep, but I know you can do it.” Similarly, rewards and praise we used to reinforce progress on sleep-related concerns (e.g., transitioning from co-sleeping to independent sleeping).

Session 3: Reduce Parental Accommodation and Avoidance. During the third session, parents were provided with psychoeducation about parental accommodation and skills for reducing child avoidance (e.g., exposure, “*The Habituation Curve*”, “*Sample Fear Hierarchy*”; (Merson & Chu, 2011)). Parents were instructed to refrain from “saving” the child from the anxiety-provoking situation and to instead provide positive reinforcement (e.g., rewards) when the child faced their fears (Kendall et al., 2005). The treatment session included a focus on reducing parental accommodation because parental accommodation is prevalent in children with anxiety (Lebowitz, Panza, & Bloch, 2016) and may reinforce avoidance and anxiety (Peterman et al., 2015; Piacentini et al., 2011; Thompson-Hollands, Kerns, Pincus, & Comer, 2014). For example, allowing the child to sleep in the parents’ bed when anxious may reinforce the child’s anxious belief that they cannot sleep alone and make the transition back to independent sleep more difficult. Exposures are one effective way to slowly reduce parental accommodation and increase the child’s confidence in their ability to overcome their fears. This session was tailored to this population by teaching parents to implement exposure while reducing parental accommodation around sleep. For example, a child afraid of going to a sleepover may slowly work up to it by practicing different components of a sleepover (e.g., sleeping in a different place in the house, sleeping in a sleeping bag).

Session 4: Maintenance and Relapse Prevention. The final session of treatment focused on reviewing skills, discussing maintenance of gains, and teaching relapse prevention techniques. For example, the therapist engaged the parent in problem solving any concerns parents had about continuing exposures, rewards, and other skills on their

own after treatment completion. Therapists instructed parents about the importance of practice for maintaining treatment gains, and also taught relapse prevention skills to facilitate continued use and generalizability of skills. These skills were tailored for this population by framing these skills in the context of sleep and anxiety. For example, we specifically included a plan for how to address sleep and sleep anxiety during the transition from summer to the school year.

Post-Treatment Assessment. A post-treatment assessment was completed approximately one week after treatment completion. During the post-treatment assessment, independent graduate student evaluators completed the ICSD Clinical Interview with parents and children separately to determine whether children met criteria for chronic insomnia and to assign a chronic insomnia clinical severity rating. Additionally, parents and children completed questionnaires on sleep, anxiety, and other related topics through REDCap and children completed a reward learning task to determine reward learning sensitivity. Families tracked sleep for one week using child actigraphy monitors, completed daily child sleep diaries, and completed the Parent-Child Sleep Interactions Scale daily.

Children and parents also completed qualitative interviews on their experience with treatment during the post-treatment assessment. Detailed semi-structured interview guides were created by the primary investigator (Lydia Chevalier) and interviews were conducted by two undergraduate research assistants. All study staff were trained in qualitative methods. The interviews included open-ended questions designed to gather parent and child participant perceptions of the assessment procedures (e.g., sleep diaries),

treatment procedures (e.g., learning to use exposures and rewards), changes in parent behaviors around sleep (e.g., use of rewards to reinforce adaptive sleep behaviors) and treatment outcomes (e.g., changes in sleep onset latency). Specifically, relevant to the current analysis, interviews included open-ended questions focused on: 1) acceptability of treatment (e.g., “What did you think of your parent coming in to learn how to help you sleep better?”), 2) effects of treatment on parent-child sleep interactions and parenting behaviors (e.g., “What changes did you make in how you respond to your child’s anxiety around sleep?”), 3) effects of treatment on sleep outcomes (e.g., “What changes have there been in your child’s sleep since starting the study?”), and 4) impact of treatment on non-sleep outcomes (e.g., “What changes in your child’s anxiety have there been since starting the study?”).

Follow-up Assessment. A follow-up assessment was completed approximately one month after treatment completion by phone. During the follow-up assessment, independent graduate student evaluators completed the ICSD Clinical Interview with parents and children separately to determine whether children met criteria for chronic insomnia and to assign a chronic insomnia clinical severity rating. Parents and children completed questionnaires on sleep, anxiety, and other related topics through REDCap. Families tracked sleep for one week using daily child sleep diaries and completed the daily Parent-Child Sleep Interactions Scale.

Measures

Eligibility Measures

Demographics and Background History Questionnaire. This questionnaire was developed to provide demographic information on children and their families. It contains questions about basic demographic information (e.g., race, ethnicity, household income), school information (e.g., grade), medical history, and parent information (e.g., age, education, relationship to child). This questionnaire was completed by parents through REDCap at the baseline assessment.

International Classification of Sleep Disorders (ICSD) Clinical Interview. A clinical interview based on the ICSD 3rd edition (American Academy of Sleep Medicine, 2014) was conducted with parents and children to systematically gather information about sleep (e.g., sleep location) and bedtime routines. In order to meet criteria for chronic insomnia, participants must: 1) report sleep initiation or maintenance problems, 2) have adequate opportunity and circumstances to sleep, and 3) experience daytime consequences at least three times a week for at least three months. For children, resistance of going to bed on an appropriate schedule or difficulty sleeping without parent or caregiver intervention is included in criterion one. The interview included questions on the potential consequences of poor sleep (e.g., “On those four nights that your child had trouble sleeping, how often did you notice that he felt sleepy the next day?”). The clinical interview was also conducted at post-treatment and follow-up assessment by independent graduate student clinician evaluators and specified sleep within the last two weeks.

Two independent graduate-clinician raters reviewed all clinical interviews at baseline, post-treatment, and follow-up assessments to determine if participants met criteria for chronic insomnia and to assign clinical severity ratings (CSRs, i.e., 0=no interference, 4=clinically significant interference, 8=life impairing interference). All scores of four or above were considered clinical. Final CSRs were calculated by averaging the two ratings. Although the clinical interview was completed separately with both parents and children, diagnoses and CSRs were determined based on parent report only. Raters had complete diagnostic agreement at baseline assessment (100%) and near-complete agreement at post-treatment (90%) and follow-up (89%). Inter-rater reliability was good for baseline CSR ($ICC=0.85$), moderate for post-treatment CSR ($ICC=0.73$), and excellent for follow-up assessment CSR ($ICC=0.97$).

Anxiety Disorders Interview Schedule Child and Parent Report (ADIS-IV-C/P). The ADIS-IV-C/P is a semi-structured interview completed with children and parents to assess for anxiety and co-morbid disorders (e.g., mood disorders, externalizing disorders; (Silverman & Albano, 1996). The ADIS-IV-CP has been demonstrated to be a reliable structured interview for diagnosing anxiety disorders in a specialized anxiety outpatient clinic (Silverman, Saavedra, & Pina, 2001). The primary investigator, who was trained to research reliability on this interview, completed the ADIS-IV-C/P with participants during the baseline assessment to determine treatment eligibility and assign appropriate anxiety and related disorder diagnoses.

Pubertal Development Scale (PDS). The PDS is a five-item self-report questionnaire that asks about pubertal changes such as growth spurts, voice changes

(boys), and menstruation (girls) to determine a child's pubertal stage (Carskadon & Acebo, 1993). A total scale is created by averaging all of the items. As puberty significantly impacts sleep (Knutson, 2005), participants in this study must have not reached puberty at the time of enrollment (i.e., have a PDS score of three or below). The PDS has been found to be both reliable and valid (Carskadon & Acebo, 1993). All participants had a score below three at baseline ($M=1.57$, $SD=0.43$) meaning that they were pre-pubertal and eligible for the study. This questionnaire was completed by children through REDCap at the baseline assessment.

Parent-Child Sleep Interaction and Parenting Measures

Parent-Child Sleep Interactions Scale (PSIS). The PSIS is a 12-item parent-report measure used to assess sleep-related parenting behaviors and interactions between parents and children around sleep (Alfano et al., 2013). This scale provides a total score as well as three subscales: sleep dependence (e.g., “my child sleeps in my room all night”), sleep reinforcement (e.g., “I praise my child for good sleep behaviors”), and sleep conflict (e.g., “my child and I argue about bedtimes/sleep schedules”). Parents completed the original questionnaire through REDCap at baseline and were asked to note how often each of these behaviors had occurred on a scale from never (0) to always/almost always (4) in the last month. Parents completed an adapted version of the questionnaire on paper every night during baseline, treatment, post-treatment, and follow-up weeks in order to provide daily sleep reinforcement, sleep dependence, and sleep conflict scores. In the daily questionnaire parents noted whether a specific behavior (e.g., “I reassured my child he/she was safe last night”) had occurred that night (1) or not (0).

Although this scale has been validated for use with children with sleep concerns (Alfano et al., 2013), it has not yet been validated in children with anxiety and SRPs.

Family Accommodation Checklist and Interference Scale (FACLIS). The FACLIS is a 20-item parent-report questionnaire designed to capture the range of different parental accommodation behaviors performed by parents, as well as the interference that these behaviors cause in the family (Thompson-Hollands et al., 2014). Parents were asked if they had engaged in a specific accommodation (e.g., “ordered for my child at a restaurant”) over the last two weeks. Number of accommodations were summed for total accommodation score; this number was used to represent the scope of parental accommodation in this study. For each accommodation endorsed, parents were asked to rate how much performing this action interfered in their life (0 = “No interference” to 8 = “Extreme interference”). An average interference score was calculated by dividing the total accommodation score (i.e., number of accommodations) by the total amount of interference. This scale has been found to be valid and reliable in a sample of treatment-seeking anxious youth (Thompson-Hollands et al., 2014). Parents completed this questionnaire through REDCap during baseline, post-treatment, and follow-up assessments.

Parenting Scale. The Parenting Scale is a 30-item parent-report measure that measures dysfunctional discipline practices of parents such as laxness, over-reactivity, and verbosity (Arnold, O'leary, Wolff, & Acker, 1993). Parents were asked to rate their behaviors in situations (e.g., “when my child misbehaves”) on a continuum from effective discipline (e.g., “I speak to my child calmly”) to ineffective discipline (e.g., “I

raise my voice or yell”). After reverse scoring several items, lower scores are associated with more effective discipline. A total score is created by averaging all items and subscales (i.e., laxness, over-reactivity, verbosity) are created by averaging items on that subscale. The Parenting Scale has demonstrated adequate internal consistency and test–retest reliability (Arnold et al., 1993). Parents completed this questionnaire through REDCap during baseline, post-treatment, and follow-up assessments.

Sleep Outcome Measures

Children’s Sleep Habits Questionnaire (CSHQ). The CSHQ is a 45-item parent questionnaire designed to assess a range of sleep-related problems (e.g., bedtime resistance, sleep anxiety, sleep onset latency) in school-aged children (Owens, Spirito, & McGuinn, 2000). Parents rated the frequency of child sleep behaviors (e.g., “child resists going to bed at bedtime”) from rarely (0-1 times a week) to typically (5-7 times a week). A cutoff total score of 41 was selected for optimal sensitivity (0.80) and specificity (0.72; (Owens et al., 2000). The CSHQ is a reliable and valid instrument for assessing sleep in clinical and non-clinical samples (Owens et al., 2000). In addition to the total score, the CSHQ also provides subscales (e.g., sleep anxiety, bedtime resistance, sleep onset latency, sleep duration, night wakings, daytime sleepiness, parasomnias) that capture specific SRPs. Parents completed this questionnaire through REDCap during baseline, post-treatment, and follow-up assessments.

Children’s Report of Sleep Patterns Questionnaire (CRSP). The CRSP is a 60-item questionnaire that assesses sleep patterns, sleep hygiene, and sleep disturbance in

school-aged children (Meltzer et al., 2013). Children estimated the frequency of sleep behaviors (e.g., “have trouble falling asleep at bedtime”) on a scale of zero (“never”) to four (“always”). In addition to providing information on sleep timing (e.g., wake time), this measure includes a variety of subscales (e.g., insomnia, bedtime fears/worries, sleep location). The CRSP demonstrates good reliability and validity (Meltzer et al., 2013). Child participants completed the CRSP through REDCap during baseline, post-treatment, and follow-up assessments.

Sleep Diaries. Sleep diaries were used to calculate subjective sleep onset latency (SOL). SOL is generally defined as the number of minutes between when a child goes to bed (i.e., bed time) and when the child falls asleep. Each day, parents and children together completed a sleep diary for the child that noted the time that the child went to bed and an estimate of when the child fell asleep. Sleep diaries are recommended to get a full understanding of sleep behaviors, and research has supported their validity and reliability (Gaina, Sekine, Chen, Hamanishi, & Kagamimori, 2004; Hudson, Gradisar, Gamble, Schniering, & Rebelo, 2009; Paine & Gradisar, 2011). Sleep diaries were completed daily on paper during baseline, treatment, post-treatment, and follow-up weeks.

Actigraphy. Actigraphy, the measurement of sleep/wake cycles through wristwatch-like devices worn 24 hours a day (Cousins et al., 2011; Paine & Gradisar, 2011), was used as the objective measure of sleep onset latency during the baseline, treatment, and post-treatment weeks of the study. Specifically, the ActiGraph GT9X Link, an actigraphy monitor commonly used to measure child sleep in school-aged

children (Brazendale et al., 2019), was used for this study. Actigraphy has been used previously to evaluate sleep of preschool children with bedtime worries (Kushnir & Sadeh, 2013) and school-aged children with generalized anxiety disorder (Alfano et al., 2015). Actigraphy data was only available for eight participants as one participant (P4) did not wear the watch reliably enough for data analysis and another (P9) did not wear a watch because all watches were in use at that time.

Non-Sleep Outcome Measures

Multidimensional Anxiety Scale for Children (MASC). The MASC is a 30-item child-report/parent-report questionnaire designed to assess various dimensions of anxiety (March, Parker, Sullivan, Stallings, & Conners, 1997). The child-report version contains items such as “I get shaky or jittery” that are rated on a four-point scale (0= “never true about me,” 3= “often true about me”). The parent-report version contains parallel items (e.g., “my child gets shaky or jittery”) rated on the same four-point scale (0= “never”, 3= “often”). Both scales produce a total score which were used to examine overall child anxiety. The MASC has good internal consistency (Cronbach’s alpha = 0.87), high test-retest reliability, and good convergent and divergent ability (March et al., 1997). Parents and children completed this questionnaire through REDCap during baseline, post-treatment, and follow-up assessments.

Pediatric Symptom Checklist (PSC). The PSC is a 35-item parent-report questionnaire designed to measure psychosocial dysfunction in school-aged children and to assess other potential areas of dysfunction (e.g., externalizing symptoms,

oppositonality) (Jellinek et al., 1988). Parents were asked to rate items about their child (e.g., “distracted easily”) as “never” (0), “sometimes” (1), or “often” (2) describing their child. A total score is calculated by adding all items and a score of 28 or higher is classified as clinically significant impairment. The Pediatric Symptom checklist has been found to be valid with a specificity of 0.68 and sensitivity of 0.95 (Jellinek et al., 1988). This questionnaire was completed by parents through REDCap during baseline, post-treatment, and follow-up assessments.

Brief Impairment Scale (BIS). The BIS is a 23-item parent-report questionnaire designed to assess general functioning in domains such as interpersonal relations, school/work functioning, and self-care/self-fulfillment (Bird et al., 2005). Parents were asked to rate how much of a problem a certain topic is (e.g., “getting along with his/her brothers and sisters”) on a scale from 0 (“not a problem”) to 3 (“a serious problem”). The BIS has been found to have good reliability and validity (Bird et al., 2005). This questionnaire was completed by parents through REDCap during baseline, post-treatment, and follow-up assessments.

Reward Learning Task. Children completed a probabilistic reward task to determine reward learning sensitivity. The task utilized in this study was developed by Pizzagalli and colleagues (Pizzagalli, Jahn, & O’Shea, 2005) and requires participants to identify whether a long or short mouth is presented on a face, after which they are rewarded with a message that reads “Correct!! You won 5 Cents.” Correct identification of one stimulus is rewarded three times more often than the other stimulus. Task performance is assessed and response bias towards the reinforced stimulus is used to

evaluate the extent to which participants are sensitive to reward learning. Participants with high reward learning capacity generally demonstrate more of a response bias because they are changing their responses based on rewards, while participants with low sensitivity to reward learning will still perform fairly well but have less of a response bias (Morris & Rottenberg, 2015). In previous studies, this reward learning task was sensitive enough to differentiate between participants with normal reward learning in GAD and reduced reward learning capacity in depression (Morris & Rottenberg, 2015). Additionally, this task appeared to demonstrate an adequate spread of reward learning sensitivity scores across participants (Morris & Rottenberg, 2015), as did a similar task used with boys with depression, anxiety, or an externalizing disorder (Forbes, Shaw, & Dahl, 2007).

Data Analytic Plan

Baseline, Post-Treatment, and Follow-up Assessment Questionnaire Data

Analyses. For questionnaire data collected during the baseline, post-treatment, and follow-up assessments, we ran descriptive statistics, correlations, and Wilcoxon signed-rank tests to examine the effects of treatment on variables of interest (e.g., frequency of sleep anxiety at post-treatment and follow-up compared to baseline). Wilcoxon signed-rank tests were used instead of paired t-tests as they can provide accurate results without requiring a normal sampling distribution, an assumption often not met when analyzing data from small samples (Ghasemi & Zahediasl, 2012; Shieh, Jan, & Randles, 2007). Effect sizes for the Wilcoxon signed-rank tests were calculated using the appropriate formula ($r = z/\sqrt{N}$; (Field, 2009), and interpreted as small (0.10), medium (0.30), or large

(0.50) effects (Cohen, 1992).

Single Case Data Analyses. Consistent with current single-case recommendations, visual analysis and the calculation of p-values and effect sizes were completed for graphs of sleep reinforcement, sleep conflict, and sleep dependence (Barlow et al., 2009; Manolov & Moeyaert, 2017). Graphs for each participant were created and the changes in level (mean of data within a phase), trend (slope), variability (deviation of data around the slope of best fit), and overlap (percentage of data from one phase that overlaps with data from another phase) were examined to determine the effect of treatment on these variables (Barlow et al., 2009; Manolov & Moeyaert, 2017). As is traditional in single case analyses, graphs are presented in order of baseline lengths (e.g., one-week baseline before two-week baseline) rather than in order of participant number.

An effect size technique called non-overlap of all pairs (NAP) was used to examine the proportion of data points in the treatment phase that were an improvement from the baseline data points (Manolov & Moeyaert, 2017; Parker & Vannest, 2009). Every pair of observations from different phases was compared (e.g., Baseline 1-Treatment 1, Baseline 1-Treatment 2). A NAP score of 0.5 corresponds to zero effect because the number of “improved” pairs is equal to “non-improved” pairs. When applicable, scores were reversed so that values of above 0.5 represent improvement (i.e., reduction in sleep dependence). Significant effect sizes were classified as weak (0-0.65), medium (0.66–0.92), or large (0.93–1.0). This method was selected because it remedies the weaknesses of more common effect sizes such as percent non-overlapping data (PND) (Hanley & McNeil, 1982; Parker & Vannest, 2009). This technique was also

selected over parametric analyses such as t-tests because data in approximately three quarters of single case designs do not meet requirements needed for parametric tests, often contain outliers which may skew results in parametric tests, and are auto-correlated, making statistical analysis more difficult (Hedges, Pustejovsky, & Shadish, 2013; Parker & Vannest, 2009).

Qualitative Data Analyses. Qualitative interviews were recorded, transcribed verbatim, and cleaned for spelling and accuracy by research assistants trained in qualitative transcribing. Interviews were then analyzed in NVivo 10 (QSR International Pty Ltd, 2012) using applied thematic analysis (Guest, MacQueen, & Namey, 2011). A preliminary coding structure was developed by a coding team of two psychology doctoral students with previous experience using applied thematic analysis and was based on a-priori research questions (see post-treatment assessment section in methods section above for a list of these research questions). The first two transcripts were coded together by the coding team in order to refine and test the coding structure. The coding structure, including the name, descriptions, and usage examples, was then updated iteratively to include themes that emerged from the data. Twenty percent of the transcripts were double coded and compared to ensure reliability across raters; inter-coder concordance (Morse, Barrett, Mayan, Olson, & Spiers, 2002) was calculated to be 87%. The remaining transcripts were coded individually and discussed at weekly meetings to resolve any questions and update the coding structure accordingly.

In the current analyses, data were first queried for all participants to assess parent and child perception of treatment (e.g., acceptability, what skills they remembered

learning). Next, data were queried for all parent participants to assess parent perception of changes in parent-child sleep interactions (e.g., skills used to reduce sleep conflict). Data for child participants were not included for this section as children did not participate in sessions and were not directly asked about different types of parent-child sleep interactions. Next, data were queried for all participants to assess parent and child perceptions of the effects of treatment on sleep outcomes (e.g., sleep onset latency). Finally, data were queried for all participants to assess parent and child perceptions of the effects of treatment on non-sleep outcomes (e.g., anxiety). The coding team collaboratively categorized similar codes within these areas into themes to most accurately illustrate the experiences of the participants.

Results

Quantitative Outcomes

Treatment Feasibility and Fidelity

All ten participants completed treatment, attending all four parent training sessions and engaging in all four check-in calls. All participants completed treatment within approximately four weeks ($M=25.8$ days, $SD=7.30$), as intended. One participant was lost to follow-up after completing the post-treatment assessment, and the remaining nine participants completed all study procedures. As suggested by Bouchtein and colleagues (2019) to evaluate treatment fidelity, two outside raters reviewed a randomly selected session for each participant (Bouchtein et al., 2019). These two trained independent evaluators watched the randomly selected session tapes and provided a coverage rating (i.e., 0=not covered, 1=superficially covered, 2=covered thoroughly) for

each element (e.g., reviewing a sleep hygiene handout). The two trained independent evaluators rated 78% of treatment elements to be covered thoroughly (i.e., discussed in detail, examples elicited from parent) and approximately 22% to be superficially covered (e.g., discussed but not applicable to child so not covered in depth). For example, if a child was already waking up at the same time every day, the clinician would briefly discuss the importance of consistent wake times, but would not cover this topic in depth because this was not an area of concern for this participant. All treatment elements were deemed to be consistent with the manual.

Effects of Treatment on Parent-Child Sleep Interactions and Parenting

Sleep Reinforcement Single Case Data. Visual analysis of the baseline sleep reinforcement data trends indicated that baseline sleep reinforcement was stable or decreasing for the majority of participants, indicating that these data were appropriate to analyze. However, baseline trends for P7, P8, and P10 were increasing, so interpretations for these participants should be made with caution as they may represent a continuation of baseline increase in sleep reinforcement. Visual analysis of sleep reinforcement data indicated change in level and slope in the desired direction (i.e., increases in sleep reinforcement) across most assessment periods for eight participants (P2, P3, P4, P5, P6, P7, P9, P10) and decreases in sleep reinforcement for two participants (P1, P8). However, only P2, P3, P4, P5, P6, and P9 demonstrated significant improvements in sleep reinforcement. Please refer to Table 3 for all NAP scores, significant values, and effect sizes and Figure 2 for sleep reinforcement single case graphs. As is traditional in single case analyses, graphs are presented in order of baseline lengths (e.g., one-week baseline

before two-week baseline) rather than in order of participant number.

Sleep Conflict Single Case Data. Visual analysis of the baseline sleep conflict data trends indicated that baseline sleep conflict was stable or increasing for the majority of participants, indicating that these data were appropriate to analyze. However, baseline trends for P1, P3, and P10 were decreasing, so interpretation for these participants should be made with caution as they may represent a continuation of baseline decrease in sleep conflict. Visual analysis of sleep conflict data indicated change in level and slope in the desired direction (i.e., decreases in sleep conflict) across most assessment periods for three participants (P1, P3, P10), no change for five participants (P4, P5, P7, P8, P9), and increases in sleep conflict for two participants (P2, P6). Only P2 and P6 demonstrated significant changes in sleep conflict, with P2 experiencing a significant increase in sleep conflict during treatment and P6 experiencing a significant increase in sleep conflict during follow-up. Please refer to Table 3 for all NAP scores, significant values, and effect sizes and Figure 3 for sleep conflict single case graphs.

Sleep Dependence Single Case Data. Visual analysis of the baseline sleep dependence data trends indicated that baseline sleep dependence was stable or increasing for the majority of participants, indicating that these data were appropriate to analyze. However, baseline trends for P3, P4, P8, and P10 were decreasing, so interpretations for these participants should be made with caution as they may represent a continuation of baseline decrease in sleep dependence. Visual analysis of sleep dependence data indicated change in level and slope in the desired direction (i.e., decreases in sleep dependence) across most assessment periods for eight participants (P2, P3, P4, P6, P7,

P8, P9, P10) and increases in sleep dependence for two participants (P1, P5). However, only two participants demonstrated significant reductions (P2, P7), and P5 had a significant increase in sleep dependence during post-treatment. Please refer to Table 3 for all NAP scores, significant values, and effect sizes and Figure 4 for sleep dependence single case graphs.

Parenting Questionnaires at Baseline, Post-Treatment, and Follow-up. Rates of parental accommodation (i.e., FACLIS total score) were high in this sample and parents reported significantly fewer accommodations at post-treatment and follow-up compared to baseline with large effect sizes (Table 4). Although not significantly different from baseline to post-treatment assessment, total interference of accommodations (i.e., FACLIS total interference score) was significantly lower at follow-up assessment than baseline with a large effect size. Rates of ineffective discipline (i.e., Parenting Scale total score) were high in this sample and significantly lower at post-treatment and follow-up than baseline with large effect sizes (Table 4). Although not statistically significant, parenting laxness, overreactivity, and verbosity were lower at post-treatment and follow-up compared to baseline. Please refer to Table 4 for all Wilcoxon signed-rank test results.

Effects of Treatment on Sleep Outcomes

Diagnostic Outcomes. At the post-treatment assessment one participant met criteria for a diagnosis of chronic insomnia and the remaining nine participants no longer met criteria for chronic insomnia. At the follow-up assessment, three participants met criteria for chronic insomnia, six did not meet criteria, and one was lost to follow-up.

Clinical severity ratings were significantly lower at post-treatment and follow-up assessments with large effect sizes according to Wilcoxon signed rank tests (Table 5). Table 1 displays diagnostic criteria (e.g., sleep dependence) and potential consequences of poor sleep (e.g., irritability) at baseline, post-treatment, and follow-up.

Sleep-Related Problem Questionnaires at Baseline, Post-Treatment and Follow-up. Total CSHQ score, CSHQ sleep onset latency, CSHQ sleep anxiety, CSHQ bedtime resistance, and CSHQ sleep duration were significantly improved at post-treatment and follow-up assessments compared to baseline and with large effect sizes (Table 5). There were no significant changes in CSHQ night waking scores at post-treatment, but night wakings were significantly less frequent at follow-up assessment than baseline with a large effect size. CSHQ daytime sleepiness was significantly less frequent at post-treatment with a large effect size, but reductions were no longer significant at follow-up assessment. There were no significant differences from baseline to post-treatment or follow-up assessment for parasomnias. Although not statistically significant, child participants reported improvements in insomnia, bedtime worries, and sleep location (i.e., more frequently sleeping in their own bed) on the CRSP from baseline to post-treatment and follow-up assessments. Table 5 displays Wilcoxon signed-rank tests and effect sizes.

Subjective Sleep Onset Latency Single Case Data. Visual analysis of the baseline sleep diary SOL trends indicated that baseline subjective SOLs were stable or increasing (i.e., SOL was getting longer) for the majority of participants, indicating that these data were appropriate to analyze. However, baseline trends for P4, P5, and P8 were

decreasing (i.e., taking less time to fall asleep), so interpretation for these participants should be made with caution as they may represent a continuation of baseline reduction in SOL and interpretations of the impact of treatment on SOL may not be accurate. Visual analysis of sleep SOL data indicated change in level and slope in the desired direction (i.e., reduction in SOL) across most time points for all participants. However, only P1, P3, P5, P6, P7, P9, P10 experienced significant improvements, and changes were not consistent across all assessment points. Two participants (P3, P5) were falling asleep significantly faster during treatment compared to baseline according to sleep diary data and these improvements continued into post-treatment and follow-up periods. Another two participants (P1, P10) were falling asleep significantly faster in treatment than baseline but improvements were lost during post-treatment or follow-up periods. Two participants (P7, P9) did not significantly improve during treatment, but were falling asleep significantly faster during post-treatment and/or follow-up. One participant (P6) was falling asleep faster during treatment and post-treatment periods but sleep onset latency was not significantly shorter than baseline at follow-up. Please refer to Table 6 for NAP scores, significance values, and effect sizes and Figure 5 for subjective SOL single case graphs.

Objective Sleep Onset Latency Single Case Data. Visual analysis of the baseline actigraphy sleep onset latency (SOL) trends indicated that baseline objective SOLs were stable or increasing (i.e., SOL was getting longer) for the majority of participants indicating that these data were appropriate to analyze. However, baseline trends for P3 and P7 were decreasing, so interpretations for these participants should be

made with caution as they may represent a continuation of baseline reduction in SOL and interpretations of the effects of treatment on objective SOL may not be accurate for these participants. Visual analysis of actigraphy SOL data indicated change in level and slope in the desired direction (i.e., reduction in SOL) during treatment for five participants (P2, P3, P5, P7, P10) and an increase in SOL for three participants (P1, P6, P8). However, only P3, P5, and P7 demonstrated significant change. According to actigraphy data, two participants (P5, P7) were falling asleep significantly faster during treatment and post-treatment periods compared to baseline. One participant (P3) was falling asleep faster at post-treatment, but not during treatment itself when compared to baseline. Please refer to Table 6 for NAP scores, significance values, and effect sizes and Figure 6 for objective SOL single case graphs.

Effects of Treatment on General Functioning and Symptoms

Anxiety Questionnaires at Baseline, Post-Treatment, and Follow-up. There were no significant changes in overall parent-reported child non-sleep anxiety (i.e., total MASC score) at post-treatment, but overall anxiety had significantly decreased compared to baseline with a large effect size at follow-up (Table 5). Parent-reported separation anxiety (i.e., MASC separation anxiety subscale) was significantly lower at post-treatment and follow-up assessments compared to baseline with large effect sizes. There were no significant differences in overall child-reported non-sleep anxiety (i.e., total child MASC score) or separation anxiety (i.e., child MASC separation anxiety subscale) from baseline to post-treatment or follow-up assessments. Table 5 displays Wilcoxon signed-rank tests and effect sizes.

General Functioning and Symptoms Questionnaires at Baseline, Post-Treatment, and Follow-up. Scores on the Pediatric Symptom Checklist were significantly lower at post-treatment and follow-up than baseline with large effect sizes (Table 5). There were no significant differences in the total Brief Impairment Scale scores from baseline to post-treatment or follow-up assessment.

Analysis of Reward Learning Sensitivity. Although reward learning sensitivity data were collected for all participants at baseline and post-treatment assessments, measurements were not reliable enough for analysis.

Qualitative Outcomes

Perceived Acceptability of Treatment

Parents Described Treatment Positively and Children Were Positive or Neutral. All parents reported that treatment was a positive experience and helpful in addressing their child's sleep concerns. For example, one mother of an eight-year-old girl described, "It was the perfect combination of holding my hand but letting me do it myself...Sleep has been an issue for me and my family for years so I've read feels like everything there is on sleep. But having concrete reasons of why and then creating those set [plans]...we are not doing this, and we are doing this, and this is what you say. That, to me, was the most powerful." Approximately half of children described the treatment as a positive experience (e.g., "I liked it. I like coming [to the clinic].") and the remaining children felt neutral about treatment or were unsure how they felt about treatment. All parents found the phone check-in calls to be helpful and encouraging, as described by one mother of a ten-year-old girl, "[Phone check-in calls] were a great supplement to the

training sessions that were live because if I didn't understand something, or if I had a follow-up question, or if we had a really rough night or two, that helped.”

When asked about barriers related to treatment, the majority of parents discussed the difficulty of only one parent attending treatment sessions. Of the eight families that only had one parent attend sessions, the majority of these parents noted that they wished that the child’s other parent had attended at least one session as well. For example, one mother of a nine-year-old girl described, “A joint parent education [session would be helpful]; take us both to task for our parenting styles, how we contribute to the problem, and how we can redirect our efforts to contributing to the solution.” Apart from the other parent being encouraged to come to sessions, the majority of parent participants denied any other changes they would make to treatment. For example, one mother of an eight-year-old boy described, “There is not [anything we would change about treatment] ...we got as much or more out of it than we expected.” Child participants did not suggest any changes when asked.

Parents Described Remembering a Range of Skills Learned, Children Emphasized Sleep Hygiene. When asked what they remembered learning during treatment, parents described being taught psychoeducation, sleep hygiene (e.g., no screens before bed), empathize and encourage statements, positive reinforcement (e.g., rewards, praise), and exposures (e.g., moving in small steps from co-sleeping to independent sleeping). Please see Table 2 for example quotes that describe each of these treatment components. Approximately half of children described learning about and working on sleep hygiene, as described by a ten-year-old girl, “My parents started to

make me always go to bed at 7:30 every night - including weekends.” A few other children discussed rewards or exposures. All parents described intending to continue using skills learned in treatment after study completion. For example, one mother of an eight-year-old girl reported, “I mean I can use this for the rest of [child's] life. She'll be like 34 and okay [child] your prize is some new throw pillows for your apartment.”

Perceived Effects of Treatment on Parent-Child Sleep Interactions and Parenting

Sleep Reinforcement: Parents Believed Increases in Their Use of Sleep Reinforcement Helped Children Overcome Sleep Anxiety and Avoidance. The qualitative findings support the quantitative result that there were significant increases in sleep reinforcement for the majority of participants and build on this finding by describing that parents linked increases in their use of sleep reinforcement (e.g., rewards) to changes in sleep outcomes. All parents reported using rewards during treatment and the majority noted that rewards were helpful in motivating their child to face their fears around sleep, particularly those related to sleep dependence. For example, as one mother of an eleven-year-old girl described, “I think you really need some kind of incentive to overcome something that is that hard, especially when you’re a child. Now in [child]’s case, she’s unusual and did say, ‘I would still do this even without the reward’ but I’m not sure she really would.” Parents reported using a range of rewards (e.g., playing a family game) to help children overcome anxiety and make steps toward sleep independence (e.g., fall asleep with brother in the room instead of mother). The majority of parents emphasized that in order to be successful rewards must be clear, carefully planned, and linked to a specific goal. Several parents specified that learning that these

qualities (e.g., planned ahead of time) make rewards fundamentally different than bribes (which may actually reinforce non-adaptive behavior) was important in their successful use of rewards to reinforce their child's adaptive sleep habits and progress.

Sleep Conflict: Parents Increased Their Use of the Empathize and Encourage Skill in Order to Prevent Sleep Conflict. Although sleep conflict was rare in this sample according to quantitative results, qualitative findings paint a more nuanced picture. The majority of parents reported that before treatment they often responded to their child's sleep anxiety more coldly and critically than they intended (i.e., sleep conflict). For example, one mother of a nine-year-old girl noted, "My husband, while very loving, can be on the very logical, 'Of course you are going to fall asleep. Like, you have literally never not fallen asleep.' He is at that end of the spectrum and imbuing that with a little bit more warmth and preceding it with some acknowledgement [was helpful in addressing anxiety]." Parents described knowing that a cold and critical response was not effective in reducing their child's sleep anxiety, but that they had become frustrated with their perceived inability to help their child sleep better. These instances most often occurred during times of heightened stress and when facing time constraints (e.g., rushed bedtime).

The majority of parents reported that learning an alternative skill, empathize and encourage, was empowering because it gave them a more effective way to prevent sleep conflict in response to sleep anxiety that was also more consistent with their parenting values. They described learning in treatment how to use this skill to acknowledge their child's fear and the distress it caused (i.e., empathize), while also letting the child know

that they fully believed in their child's ability to overcome it (i.e., encourage). Parents described starting off using empathize and encourage to respond to their child's sleep anxiety, but then also starting to use it with their child's other symptoms (e.g., anger) in addition to with other family members (e.g., siblings, spouses). For example, one mother of a nine-year-old girl described "[Child will] sometimes say 'I am so sad' and [now I try] to not get into a big heart to heart about that; you know, to kind of like empathize and redirect with those." The qualitative findings also help to explain the increases in sleep conflict for some participants during the study as parents described that conflict often increased as they removed accommodations related to sleep dependence and then sleep conflict decreased as children became more successful at sleeping independently.

Sleep Dependence: Parents Described Reductions in Sleep Dependence That Were Not Captured by Quantitative Measures. The qualitative findings build on and help to explain the mixed quantitative improvements in sleep dependence. During qualitative interviews, the majority of parents described instances of sleep dependence that were completely resolved during treatment. For example, one mother of a nine-year-old girl described, "Prior to the study, most nights, I would lay down in her sister's bed because her sister is older and has a later bedtime until [child] fell asleep and we -- that extinguished right away." The remaining families reported some but not complete improvement, as reported by one mother of an eight-year-old girl, "She still needs like somebody. Like if I don't lay with her, she has her brother." These qualitative findings fill an important gap because many of the examples of sleep dependence described by parents during the interviews (e.g., laying with the child until she was asleep, checking

repeatedly on the child after bedtime), were not captured by the daily Parent-Child Sleep Interactions (PSIS) checklist.

Parents described engaging in these patterns of behavior in an effort to protect or save their child from anxiety or to prevent the consequences of poor sleep (e.g., co-sleeping before the child had a big test). The majority of parents described behaviors related to sleep dependence (e.g., checking on the child) as taking up a significant amount of parent time and energy, and were eager to shift them. Some parents described that children often initially disagreed with the need for change in sleep dependence (e.g., “She didn’t understand why I couldn’t sleep in the bed with her anymore, hold her hand, that type of stuff...I think she thought it was like a punishment.”), but many parents noted that their child was proud of changes they made towards sleep independence during treatment. For example, one mother of an eleven-year-old girl described her daughter’s reaction to working through difficulties with sleep dependence, “I mean, it’s changed [child’s] life...She feels so much more confident because she knows that she faced something that was really scary for her.”

Parenting: Parents Changed their Parenting Behaviors and Began Using Skills Learned in Treatment with Other Family Members. All parent participants described making changes to their parenting behaviors during the study and reported using the skills learned in treatment with other family members. The majority of parents noted using skills with their other children, but several also used skills with their partners. For example, one mother described, “If you empathize and encourage everyone around you...[it] just spills over into other relationships in a good way. I think it's just increased

patience and compassion and um, changing the way I address anxiety and [other concerns].” More specifically, the majority of parents reported using positive reinforcement such as rewards, approximately half reported using empathize and encourage, and approximately half reported using sleep hygiene techniques learned during treatment with their other children (e.g., “[Now] nobody can have iPads after seven.”). The majority of parents reported that using skills learned in treatment with their other children was very helpful. For example, one mother described, “I’m giving [sibling] rewards now and it’s really helping...[sibling] feels like she has a half therapist. She’s like, ‘Oh, what? We’re not gonna have [therapist] anymore? I love [therapist] even though I never met her.’”

Perceived Effects of Treatment on Sleep Outcomes

The majority of parents and children reported that child sleep had improved over the course of treatment. Parents described changes in child sleep onset latency and sleep dependence as the most meaningful sleep outcomes of treatment. For example, one father of an eight-year-old girl reported, “[Child] doesn’t come out in the middle of the night and kick me out the bed...or come out early like at nine and say I can’t sleep, I can’t sleep, what am I going to do at school?...I mean that’s giant.” Children described improvements in sleep onset latency and sleep dependence as well, but also emphasized improvements in daytime tiredness. Approximately half of children reported feeling less tired during the day and having more energy. For example, an eight-year-old girl reported, “At first I was super tired because I couldn’t...fall asleep and I stayed up until like ten thirty, but now because I go to sleep right away, I’m not tired anymore.” Some

children also described changes in their energy and how they felt when they woke up after treatment as compared to before. For example, a ten-year-old girl described, “I’m not as tired in the day...I feel much more energetic and I don’t feel droopy. And when I wake up, I’m not all groggy. I get up and I’m like ‘okay, so let’s get dressed!’.” Parents and children reported that wake after sleep onset had never been a concern and was not generally addressed in treatment (e.g., “I just have trouble falling asleep but once I’m asleep, I don’t need to get up or anything like that.”). Sleep duration and sleep quality were rarely discussed by parents or children during interviews.

Sleep Anxiety: Worries About Not Being Able to Sleep Decreased During Treatment but Worries About Separation at Bedtime Increased Before Decreasing.

Consistent with the quantitative CSHQ results, the majority of parents reported overall improvements in sleep anxiety. The qualitative findings add to these results by providing additional information on the different types of sleep anxiety in this sample and the ways in which sleep anxiety changed across treatment. The majority of parents reported that before treatment their child was anxious about not being able to fall asleep or the consequences of poor sleep. For example, one mother of an eleven-year-old girl noted, “[Child was worried] that she couldn’t fall asleep and that she would not be okay the next day, that she would be a wreck”. These parents described meaningful reductions in this type of sleep anxiety at the post-treatment assessment. For example, one mother of a ten-year-old girl described, “[Child used to say] ‘I can’t fall asleep, what’s wrong with me? I need help, I’m broken. I’m going to be sick, I’m going to make myself sick.’...those are big statements and those have ended now.” Many parents reported that as children began

to be more successful in falling asleep quickly, they experienced increased confidence in their ability to sleep and to handle potential consequences of poor sleep.

Approximately half of parents reported that their child was afraid of separating from parents or other family members at night in addition to fears of not being able to fall asleep. For example, one mother of an eight-year-old girl noted, “I think it was separation a little bit. She wanted to be with us. Um and the fear that she wouldn't get to sleep and you know, get enough sleep.” Parents of these children described that the child experienced increases in anxiety as they practiced relying less on parents at bedtime and then drops in anxiety as they adjusted to this new sleep independence. Child participants rarely discussed sleep anxiety, but several noted that they experienced anxiety about separating from parents around sleep and that this anxiety had improved during treatment. For example, an eleven-year-old girl reported, “I don't like [practicing sleepovers] because it makes me anxious...but I do like that's it's actually, it definitely is helping me.”

Sleep Onset Latency: Parents and Children Reported Reductions in Sleep Onset Latency that Some Parents Linked to Changes in Sleep Misperception.

Consistent with the quantitative (sleep diary) findings of reductions in subjective sleep onset latency, the majority of parents felt that their child was falling asleep faster after treatment. For example, one mother of an eight-year-old girl reported, “[Child] was in bed awake for hours and hours before and now she's asleep earlier.” Several parents reported that child sleep onset latency initially worsened during treatment, especially as they worked towards sleep independence, and then improved as children became more

confident in their abilities to sleep independently. Similarly, the majority of child participants believed that their sleep improved, but several reported that their sleep stayed the same during treatment. Approximately half of child participants specified that they were falling asleep faster after treatment, as an eleven-year-old girl described, “I think it’s been like a lot better, because I’m falling asleep more quickly and there’s been a lot less of uh, staying up late.” The others either believed that they were taking the same amount of time to fall asleep or did not specifically mention sleep onset latency when discussing improvements in sleep.

The qualitative findings help to explain the discrepancy between the subjective and objective SOL quantitative results by demonstrating that some of the change in subjective SOL may have been due to changes in sleep misperception. Some parents realized that they and their child had developed misperceptions about sleep, and discovered that their child was actually sleeping much better than they had believed. For example, one mother of a ten-year-old girl reported, “The data itself sometimes surprised me that she was in fact falling asleep quicker than we thought and I think [child] was thinking she was lying there longer than she was.” Using information learned in treatment (e.g., sleep psychoeducation) these parents were able to directly address this sleep misperception by providing their child with more accurate information about their sleep. Parents of children without sleep misperception also used this technique over the study to help their child see the progress they were making in treatment. For example, one mother of a nine-year-old girl reported, “I pointed out to her that she’s never had a night where she’s literally never fallen asleep...and as the study went on and her sleep actually got

better, I was able to point that out to her and help her reality check.”

Perceived Effects of Treatment on General Functioning and Symptoms

Anxiety: Non-Sleep Anxiety Reduced but Not Completely Resolved During Treatment. The qualitative findings build on the quantitative finding (parent-report MASC) that non-sleep anxiety was not reduced until follow-up by showing that as treatment progressed parents started using the skills on non-sleep anxiety. For example, in the same way that parents learned how to not accidentally agree with or reinforce sleep anxiety and avoidance, many mentioned working on not reinforcing other types of anxiety. Despite these improvements and use of skills, the majority of parents reported remaining non-sleep anxiety after treatment and many parents were unsure of how to most effectively apply the skills to these non-sleep worries. For example, one mother of an eight-year-old girl described, “She still has different kinds of areas of anxiety that I feel like I don't know how to work on um that sort of aren't as conducive to the [reward] system...but for sleep her anxiety is better and she has tools for that.”

General Functioning: Parents Reported Most Improvements in Family Functioning, While Children Reported More Improvements in School Functioning. Overall, parents reported that children were generally able to function well academically and socially and that most of the consequences of poor sleep before treatment occurred at home. For this reason, parents reported that they generally did not see large improvements at school as children were already doing well. For example, the mother of a ten-year-old girl described, “[Child’s] sleep difficulties never manifested in other activities; I never heard from school; I never heard from the piano teacher, tennis,

friends...In our house, she would be...inconsolable about what [not being able to sleep] meant for her.” However, the majority of children reported improvements at school, mostly related to changes in daytime sleepiness and ability to focus. For example, a ten-year-old girl described, “At school when I used to not go to sleep really well, I’d be kinda groggy and sometimes...I couldn’t listen that well. So now that makes it easier, and I’m ready for school.”

Parents and children were generally in agreement that treatment did not change children’s social functioning because consequences of poor sleep (e.g., irritability) were generally confined to family rather than peers. The majority of parents and children reported improvements in family functioning (e.g., reduced conflict) after treatment. Some parents noted that as their child needed less support around bedtime they were able to spend more quality time with their other children or partners. For example, the mother of a nine-year-old girl described, “I am a bit more available in the evenings [now that child is more independent around sleep]. So my husband and I have more time to talk, I have more of a one-on-one time with my older daughter.”

Discussion

The findings of the current study support the feasibility, acceptability, and efficacy of a brief, four-session parent training intervention to treat SRPs in school-aged children with anxiety. More specifically, the four-session treatment was implemented with all participants with high fidelity to the manual, was acceptable to participants, and appeared to be effective, as children’s sleep-related problems were largely resolved after treatment. This is particularly important as SRPs are highly prevalent in school-aged

children with anxiety, and are connected with significant mental and physical health consequences (Alfano et al., 2007; Busch et al., 2017), but little research has examined sleep treatments designed specifically for this population and tailored to this age group (Bourchtein et al., 2019; Busch et al., 2017; Meltzer, 2017). Although future studies, particularly those with longer length follow-up periods, are required to provide corroboration, the findings from the present study provide support that even long-standing SRPs might be improved with a brief, intensive, parent-focused treatment as the majority of participants in the study had been suffering from SRPs for over four years, and many for their entire life.

In addition to being effective in improving sleep, the parent training intervention in this study was felt to be appropriate and acceptable to participants as feedback about the treatment was overwhelmingly positive according to parent report during qualitative interviews. For example, parents described the powerful relief of finally having effective ways to help their child sleep better after struggling with their child's sleep for years. Parents generally did not suggest any changes to the treatment itself, but the majority wished that the child's other parent had attended sessions as well so that the learning and implementation of skills could be shared between the parents. Although children did not attend sessions, their evaluation of the treatment was either positive or neutral, and they did not suggest any changes to the treatment. Although some of these positive responses were likely influenced by social desirability, all of the parents and the majority of the children provided specific instances in which skills learned in treatment had been instrumental in improving child sleep.

Effects of Treatment on Sleep Outcomes

The current study demonstrated that the parent training intervention was effective in treating SRPs in this sample, as the majority of participants no longer met criteria for chronic insomnia at post-treatment and follow-up assessments according to parent report. Consistent with previous research on SRPs in children with anxiety (Alfano, 2018), participants in this study most often experienced SRPs related to sleep dependence, sleep anxiety, and longer subjective sleep onset latency. Findings demonstrate that the intervention may be effective in addressing these SRPs as parents reported on questionnaires and during qualitative interviews that SRPs were significantly less frequent after treatment. However, although children reported insomnia and bedtime worries more frequently than a similar sample of school-aged children with clinical sleep disturbances (Meltzer et al., 2013) and noted reductions in SRPs at post-treatment and follow-up, these changes were not significant. This discrepancy is likely due to the fact that children with anxiety and their parents often report significantly different patterns of child sleep (Alfano, 2018). In this sample, for example, many participants met criteria for chronic insomnia due to parent report of sleep dependence, an issue children may not believe to be problematic (Alfano, 2018). Indeed, although parents reported significant difficulties with sleep dependence at baseline, child report on the similar metric of sleep location (i.e., frequency of sleeping somewhere other than the child's own bed) indicated that children did not find sleeping with a parent to be problematic. As much of the treatment focus and effects were related to sleep dependence, children would be expected to report less significant change during treatment if they did not see sleep dependence to

be a problem at baseline.

Qualitative findings help to illustrate and expand on the mixed parent- and child-reported changes in SRPs during the study. Despite the lack of significant change in SRPs reported by children on questionnaires, the majority of children and parents reported during qualitative interviews that child sleep had improved during treatment. In addition to changes in sleep dependence and sleep onset latency that will be discussed in later sections, children in particular emphasized improvements in daytime sleepiness that they linked to better school functioning. Although parents rarely discussed changes in their child's daytime sleepiness, the majority reported that changes in their child's sleep anxiety was one of the most meaningful improvements in SRPs that they saw during treatment. As hypothesized, parents reported decreases in child sleep anxiety related to not being able to fall asleep, however, they reported inconsistent improvements in sleep anxiety related to sleep dependence. Specifically, parents described that their child's anxiety initially *increased* when they started working towards sleep independence, but that this anxiety then dropped when their child became more confident in their ability to sleep without parental assistance.

In addition to showing a successful reduction in the frequency of SRPs after treatment, the current study helps to elucidate some of the mixed findings in the literature on subjective and objective sleep onset latency (SOL) in children with anxiety (Alfano, 2018). Although the majority of participants reported significant reductions in subjective (i.e., sleep diary) sleep onset latency (SOL) and noted improved perceived consequences of poor sleep (e.g., irritability) after treatment, there were not significant reductions in

objective (i.e., actigraphy) SOL for the majority of participants. This is consistent with several previous studies—one that also failed to demonstrate significant changes in objective sleep onset latency in a sample of adolescents with anxiety (Paavonen et al., 2016) and another that failed to demonstrate significant changes in objective sleep measures in adolescents with physical or mental health conditions (Palermo et al., 2017). The lack of significant change observed in objective SOL in the present study may be due to a ceiling effect, as few participants had clinical levels of objective SOL based on actigraphy at baseline. However, this does not explain why significant reductions in SOL were observed via subjective reports.

One potential explanation for the discrepancy between change in objective and subjective SOL is that children with anxiety and their parents may have interpreted the time between going to bed and falling asleep as longer in duration than it actually was. In fact, research shows that child anxiety impacts parent reports of SOL but not actigraphy SOL (Fletcher et al., 2018), suggesting that anxiety may impact the perception of sleep instead of the objective measure of sleep. Previous research also suggests that SRPs in children with anxiety may be the result of cognitive affective biases, low sleep self-efficacy, or sleep state misperception (e.g., believing themselves to be awake when actually asleep) rather than objective sleep disruptions (Alfano et al., 2015). In the current study, as parents and children completed these sleep diaries together and time asleep was an estimate, the subjective SOLs are likely to be influenced by shifts in sleep anxiety over time. Indeed, the majority of participants in this study experienced sleep anxiety at baseline which was significantly improved at post-treatment and follow-up according to

parent report.

Qualitative findings support the idea that improvements in subjective SOL may have been related to decreases in sleep misperception and sleep anxiety during treatment. Several parents described that before starting treatment they and their child had believed that it was taking longer for the child to fall asleep than it actually was. In fact, parents described that providing their child with more accurate information about how long it took them to fall asleep was an important component in addressing their child's sleep problems in treatment. Parents also described that their child's self-efficacy around being able to fall asleep increased during treatment, which they believed helped their child fall asleep faster. It is possible that increases in child sleep self-efficacy and reductions in sleep anxiety might help account for why subjective SOL improved but objective SOL generally did not. As children became more confident in their ability to fall asleep and anxiety related to sleep was reduced, they may have started to perceive the time between bedtime and falling asleep as subjectively shorter. Similarly, parents may have changed their perception of how long it takes their child to fall asleep both in response to information they gathered themselves (e.g., checking on the child and finding she is asleep after 30 minutes) as well as changes in how the child talked about sleep (e.g., no longer crying and saying she will never be able to fall asleep). Future research is necessary to fully understand the mechanisms of action that might be contributing to these improvements, but it is possible that for some children with anxiety, improvement is due to a shift in the *perception* of SOL rather than an *actual* change in objective SOL. Although research is very limited in children, previous studies with adults have

demonstrated that reversing misperception of sleep improves sleep outcomes (Harvey & Tang, 2012).

The lack of change in objective SOL is all the more interesting considering both the presence of significant consequences of poor sleep (e.g., irritability, impaired functioning) at baseline and the reported improvement of these consequences over treatment. Although the consequences of sleep problems are significant and clear overall (Beebe, 2011; Chaput et al., 2016; Shanahan et al., 2014; Smaldone et al., 2009; Vriend et al., 2013), there is limited research on the consequences of sleep problems in children with anxiety specifically (Alfano et al., 2007), and we are not aware of any studies that examine changes in the consequences of sleep problems over time. In the current study, all participants, even those with normal SOLs according to actigraphy, endorsed consequences of poor sleep at least three days a week. The most common negative consequences of poor sleep were increased irritability, impaired attention, sleep anxiety, and impaired social/family/academic functioning. This suggests that the perception of poor sleep may result in perceived daily consequences despite normal objective sleep. It is possible that this finding is due to objective SOL not fully capturing poor sleep in these individuals. However, as sleep metrics other than SOL (e.g., total sleep time) are generally no different in children with anxiety compared to healthy controls (Alfano, 2018), and participants in the current study rarely experienced other types of sleep difficulties at baseline (e.g., difficulties with wake after sleep onset), it may be that the perception of poor sleep is in fact linked to negative daytime consequences of poor sleep.

In adults, discrepancies between subjective and objective sleep estimates have real life consequences, and sleep treatments are effective in addressing these discrepancies (Crönlein et al., 2019). For example, adults with insomnia often overestimate their SOL compared to objective measurements and experience real consequences such as daytime sleepiness, poorer cognitive functioning, and greater functional disability (Harvey & Tang, 2012). In fact, in some cases, adults who perceive their sleep to be poor do worse on tasks than adults with objective sleep difficulties. One study found that adults who were told that they slept poorly experienced significantly more impaired daytime functioning (e.g., sleepiness), than those who were told their sleep was good despite having the same objective sleep (Semler & Harvey, 2005). This is also important because adults who perceive themselves to be poor sleepers despite normal sleep often become so anxious about not being able to sleep that they cause objective sleep problems later on (Harvey & Tang, 2012). Screening designed to identify children in the early stages of sleep misperception may be a first step toward reducing sleep anxiety and preventing the development of more significant objective sleep difficulties. The current study demonstrates that it is possible change one's perception of SOL with brief treatment and in doing so reduce the negative effects of SRPs and anxiety as well as improve overall functioning.

Effects of Treatment on General Functioning and Symptoms

Along with changes in sleep anxiety and other SRPs, parents reported improvements in non-sleep symptoms and overall functioning. For example, parents reported that children's anxiety unrelated to sleep ("non-sleep anxiety") significantly

decreased with a large effect size at follow-up compared to baseline assessment, although there was no significant difference between post-treatment and baseline. The qualitative results provide a possible explanation as to why non-sleep anxiety was not significantly reduced until follow-up. Parents described starting to apply the skills (e.g., rewards, exposures) to “non-sleep” areas of anxiety as treatment progressed, particularly in the 4th week of treatment and during the post-treatment period (i.e., week between treatment completion and the post-treatment assessment). As parents generally did not start to apply the skills to non-sleep anxiety and other symptoms until later in the study, delays would also be expected in anxiety reduction. On the other hand, separation anxiety was significantly improved at both post-treatment and follow-up by parent report, likely because the focus on reducing sleep dependence also directly addressed separation anxiety during treatment. These findings suggest that parents are able to take skills used to reduce SRPs in children with anxiety (e.g., exposures to reduce sleep dependence) and use them to successfully reduce non-sleep anxiety. However, as the majority of parents also reported being unsure of how to best apply the skills to other types of anxiety and noted that non-sleep anxiety was not completely resolved at post-treatment, additional sessions assisting parents in the use of these skills (e.g., exposures, rewards) with a wider range of anxiety presentations appears necessary.

This study suggests that a brief intervention for SRPs in children with anxiety may also generalize to improve general child functioning and other symptoms, although the quantitative results were somewhat mixed. Specifically, there were significant improvements in general functioning according to parent-report on the Pediatric

Symptom Checklist (PSC) (Jellinek et al., 1988) but not the Brief Impairment Scale (BIS) (Bird et al., 2005). In addition to the limits of small sample size to identify changes from baseline to post-treatment and follow-up, it is possible that this discrepancy was related to the items on each checklist. For example, the PSC includes items on a wide range of topics such as internalizing symptoms, externalizing symptoms, relationships, and attention, while the BIS largely focuses on how the child gets along with parents, siblings, teachers, and friends as well as behavioral problems at school and home. The BIS may have been unable to detect changes that would be applicable to this population as parents during qualitative interviews denied meaningful changes in functioning related to school or friendships as they felt their children were already doing well in these areas. Instead, parents emphasized changes in family functioning such as the parent having more time to spend with siblings or the other parent after sleep dependence had reduced. Interestingly, children did highlight improvements in school functioning, particularly related to feeling less tired and having an improved ability to pay attention at school. As the current study did not include a child-report measure of general functioning, future research should make sure to include both parent- and child-report measures of functioning to more fully understand the impact of brief sleep treatments on other domains. Future research should also examine whether changes in general functioning are related to changes in sleep, changes in anxiety, changes in parenting behaviors, or changes in other constructs.

Treatment Design and Process

Given that there is evidence from this small sample that a brief parent training intervention may be effective in reducing SRPs and anxiety as well as other non-sleep symptoms, an important next step is understanding how the intervention was designed and why it was effective. The parent training intervention developed for this study was based on general treatments for SRPs in children (Allen et al., 2016), but adapted for an anxious population with the inclusion of anxiety treatment components (e.g., rewards, exposures; (Merson & Chu, 2011)). For example, although sleep hygiene was assessed and reinforced, many families were already adhering to most general sleep recommendations (e.g., consistent wake times, dark and quiet sleep environment). Given that children with anxiety tend to have SRPs such as sleep dependence and sleep anxiety rather than sleep maintenance or quality (Alfano, 2018), components of treatment that seemed particularly relevant to anxiety and sleep (e.g., decreasing sleep anxiety, increasing sleep independence) were emphasized in this treatment for children with anxiety and SRPs.

It was clear from qualitative interviews with parents that the intended focus on sleep and anxiety treatment components relevant to SRPs in children with anxiety was appropriate and effective. For example, when asked what they remembered learning, parents accurately described psychoeducation, sleep hygiene, empathize and encourage, positive reinforcement, and exposures without prompting. Although children were not present during treatment sessions, approximately half were able to describe sleep hygiene techniques that they had learned from their parents during treatment. The fact that

children described sleep hygiene techniques, rather than other skills, may be due to the fact that sleep hygiene is a set of concrete skills and most directly applicable to why children had enrolled in treatment. Additionally, the other skills were directed more at helping parents change their behaviors around their child's sleep and children may not have been aware of their parents' use of these skills (e.g., children may not have noticed their parent using empathize and encourage statements).

Changes in Parent-Child Sleep Interactions and Parenting

As hypothesized, quantitative and qualitative results both demonstrated these treatment components were collectively effective in helping parents change their behaviors and interactions with their children around sleep. According to the single case data analyses, parents significantly increased their use of sleep reinforcement during treatment. This result was supported by the qualitative finding that parents began using sleep reinforcement skills such as rewarding and praising adaptive sleep behaviors during treatment to help children overcome their fears related to sleep. Parents noted that reinforcing adaptive sleep behaviors was particularly important in helping children shift towards sleep independence. Rewards are a common component of anxiety treatment (Kendall et al., 2005) but not necessarily part of traditional sleep interventions for children (Allen et al., 2016). As so much of the interference and consequences of SRPs for children with anxiety are related to sleep anxiety and low sleep self-efficacy, (Alfano et al., 2015), rewards and other forms of reinforcement (e.g., praise) may be an important tool in helping to reinforce independent sleep behaviors. This is particularly true for this population of anxious youth because children with anxiety often do not see their

behaviors (e.g., co-sleeping) as problematic and in need of change (Alfano, 2018). Using positive reinforcement of adaptive and independent sleep behaviors may also help children have more positive perceptions of bedtime, an important component in treating SRPs (Allen et al., 2016). Future research should examine whether rewards are particularly relevant for youth with anxiety, and/or whether they are an important element to add to treatments for children without anxiety who have SRPs.

Due to the mismatch between parent and child perception of what is problematic about sleep habits and the high rates of bedtime resistance in this population (Alfano, 2018), sleep conflict may often arise at bedtime. In the current study, contrary to hypotheses, sleep conflict was generally low at baseline and did not significantly decrease during treatment according to single case data. In fact, the only two participants to experience significant changes in sleep conflict experienced significant increases during the study. The qualitative findings help to shed light on this unexpected result as parents described sleep conflict emerging when they responded to their child's sleep anxiety in a critical manner (e.g., dismissing the child's fears about bedtime) that was not effectively captured by the Parent-Child Sleep Interaction Scale (PSIS; (Alfano et al., 2013)). Although responding in this way and engaging in sleep conflict is a common response to children's bedtime resistance, sleep conflict has been linked to poorer child sleep (Kelly, Marks, & El-Sheikh, 2014). In the case of children with anxiety and SRPs, sleep conflict likely contributes to difficulties with sleep onset latency as well as unintentionally reinforces bedtime resistance. Indeed, parents reported that they knew that this response was not effective as it increased sleep anxiety and made it more

difficult for their child to fall asleep, but this response stemmed from parents' reportedly feeling as though they were unable to help their child sleep better. Parents described that learning during treatment to respond to children by acknowledging their child's distress while also encouraging them to engage in adaptive sleep behaviors (e.g., empathize and encourage), was one of the most powerful tools they learned to avoid sleep conflict.

Much of the sleep conflict described by participants in this study was related to sleep dependence, the reliance on parental support to sleep that parents felt was either no longer developmentally appropriate or was interfering in family functioning. Contrary to hypotheses, sleep dependence did not significantly improve during treatment according to single case data, but qualitative findings help to explain these unexpected quantitative results. First, the qualitative results provide more nuance to the definition of sleep dependence than is possible to capture in quantitative surveys. Parents described everything from a child moving to her parent's bed every night and causing her father to sleep on the couch, to a child needing parents to stay in their own bedroom so that she knew exactly where they were when she fell asleep. This is important because questionnaires may miss some of these more complex examples of sleep dependence and often only query about co-sleeping. Without the qualitative data, many of the changes described as most meaningful by parents and most relevant to SRPs in children with anxiety (Alfano et al., 2015; Alfano, 2018; Cousins et al., 2011; Fletcher et al., 2018; Forbes et al., 2008) would not have been captured. Future research should involve developing and testing a questionnaire designed to capture the full range of parent-child sleep interactions common in children with anxiety as well as daily changes in these

interactions.

The qualitative findings also provide some insight into how improvements in these varied types of sleep dependence were made. During the qualitative interviews, all but one parent reported that their child struggled with sleep dependence at baseline and the majority noted complete improvement in this area by post-treatment. For example, at baseline one of the participants was co-sleeping due to separation anxiety which caused significant family and marriage interference (i.e., her father was sleeping on the couch every night). Throughout the study her parents helped her to slowly shift towards independence (e.g., coming into parents' bed at 4am, then 5am, then 6am) and by the end of treatment she was sleeping independently in her bed for the entire night every night.

Qualitative findings from this family and the majority of other families interviewed also help to demonstrate the ways that child SRPs, particularly sleep dependence, impact parent and family functioning (e.g., parent unable to spend time with other children at night due to child's sleep anxiety). In addition to freeing up parent time, shifts to more independent sleeping may have also resulted in improvements in other areas of sleep, as parental presence at bedtime has been linked to poorer sleep for children (Allen et al., 2016). However, it is important to note that sleep dependence (e.g., co-sleeping) generally only negatively impacts child sleep and family functioning when it is in response to symptoms like child anxiety; sleep dependence that is a cultural norm and applicable to all children does not have the same detrimental effect (Mileva-Seitz, Bakermans-Kranenburg, Battaini, & Luijk, 2017). For this reason, it is important for clinicians to determine whether a child's sleep dependence is related to anxiety and

interfering in child or family functioning before intervening.

In addition to shifts in parent-child interactions such as sleep reinforcement, sleep conflict, and sleep dependence, findings suggest that there were also more general changes in the ways that parents responded to their children beyond bedtime. Specifically, at baseline, rates of parental accommodation were consistent with those in other anxious samples (Kagan, Peterman, Carper, & Kendall, 2016), and were significantly lower at post-treatment and follow-up with large effect sizes. It is possible that changes in parent behaviors around sleep (e.g., reduced accommodation, reinforcement of adaptive strategies) spread to cause more general reductions in parental accommodation. For example, a parent who stops accidentally reinforcing sleep and separation fears by rewarding independent sleeping may also reinforce her child's attempts at overcoming anxiety in order to make new friends. Indeed, qualitative findings suggest that parents started to apply the skills learned in treatment to their child's other symptoms (e.g., anger) as well as with their child's siblings. Similarly, consistent with previous literature in children with anxiety (Laskey & Cartwright-Hatton, 2009), rates of dysfunctional parenting were high in this sample at baseline and were significantly lower at post-treatment and follow-up. Generally, higher parent functioning such as higher warmth, lower stress, and more consistent parenting is related to better objective and subjective sleep in children (Allen et al., 2016), so these shifts during treatment may account for some of the changes in child SRPs observed in this study.

Limitations

Although the study adds to the literature in a number of ways, the findings must be interpreted in the context of several methodological limitations. The most prominent concern is that the sample is small, so conclusions drawn from questionnaire data collected at baseline, post-treatment, and follow-up assessments (e.g., CSHQ, CRSP, MASC) should be made with caution. Although appropriate statistical methods were used (Ghasemi & Zahediasl, 2012; Shieh et al., 2007), changes in symptoms may not have been captured due to insufficient power and may not be representative of the larger population of school-aged children with anxiety. However, the use of the multiple baseline design addresses concerns about a small sample size for all of the analyses that involve single case data (e.g., subjective SOL, objective SOL, sleep dependence; Barlow et al., 2009; Rhoda et al., 2011). As a multiple baseline design was used, each participant's baseline acted as their own control condition, making much smaller samples adequate to examine the effects of treatment. More specifically, this design controlled for inter-subject variability and most threats to internal validity such as the effect of a large outside event. Additionally, the sample size of ten participants in this study is considered appropriate for a multiple baseline with three different baselines (Barlow et al., 2009) and is significantly higher than the modal number of three to four participants in most multiple baseline studies (Shadish & Sullivan, 2011).

Due to the strengths of the multiple baseline design, analyses using single case data are considered adequate to determine the effects of treatment for the individuals in the study (Barlow et al., 2009), but generalizability remains very limited. Similarly, the

qualitative methods add significant depth of understanding to the findings for the process and effects of treatment in this sample, but findings may not be generalizable to other samples. Generalizability is particularly limited for anyone other than middle class, non-Hispanic, White individuals. For example, as Black and Hispanic children have been reported to have worse sleep than non-Hispanic White children (Guglielmo, Gazmararian, Chung, Rogers, & Hale, 2018), future studies should include a more diverse sample to evaluate whether conclusions drawn from this sample are applicable to more diverse populations of youth. Similarly, as parenting around sleep (e.g., co-sleeping) is highly influenced by culture (Mileva-Seitz et al., 2017), sleep dependence may not be as much of a concern to parents in some cultures and thus treatment might focus on different SRPs in other samples.

Another limitation of the study is the fact that the self-report questionnaires selected did not fully capture the range and daily change in parenting behaviors and parent-child sleep behaviors. For example, during qualitative interviews parents described a large range of sleep dependence and sleep conflict instances that were interfering in child sleep and family functioning that were not captured on the Parent-Child Sleep Interactions (PSIS) questionnaire (Alfano et al., 2013). This is likely due to the fact that the PSIS has not been validated in an anxious sample and many of the SRPs common in children with anxiety (e.g., sleep dependence) may be much less prevalent in other populations. Future research should also examine the use of this scale in anxious populations as some parent behaviors that may be adaptive for use with non-anxious children may be problematic with anxious children. Additionally, as the PSIS was

developed to capture parent-child interactions over the past month and was adapted to daily use for this study, it was not designed to capture small daily changes in behavior (e.g., slowing transitioning from co-sleeping to independent sleeping). Future research could focus on the development of a questionnaire measure that is sensitive enough to capture changes in daily parent-child interactions around sleep in children with anxiety.

In addition to capturing outcomes, we originally included the PSIS in the study to examine mechanisms of change during treatment, but this was not possible due to several factors. For example, we planned to examine whether changes in parent behaviors (e.g., decrease in sleep conflict) preceded changes in SRPs (e.g., reduction of sleep onset latency). However, as the PSIS did not capture the majority of the instances of sleep conflict or sleep dependence, temporal and causal analyses were not possible. We had also intended on using the PSIS to examine daily changes in parent behaviors as a potential mechanism of action. Although the sleep reinforcement and sleep conflict items were framed as parent behaviors on the PSIS (e.g., “I praise my child for good sleep behaviors), sleep dependence items were more focused on the outcome (e.g., “My child sleeps in my room all night”), so it was impossible to differentiate between child and parent behaviors. For example, the PSIS could not capture if a child tried to co-sleep but the parent refused; it only captured the end result (e.g., where the child ultimately slept). The qualitative interview data demonstrated that the distinction between child and parent behaviors is important (e.g., parent sets limits on sleep dependence but child goes into parents’ bed after they fall asleep). Thus, future research should include a daily questionnaire that includes both items that capture parent behavior (e.g., parent puts child

to sleep and walks out of room) and items that capture child behaviors (e.g., child comes out of their bedroom every ten minutes after bedtime) so that these two variables can be differentiated and mechanisms can be more clearly understood. Similarly, a reward learning task (Pizzagalli, Bogdan, Ratner, & Jahn, 2007) was included at baseline and post-treatment assessments to examine whether changes in reward learning were a mechanism of treatment or whether level of reward learning sensitivity would predict response to treatment. However, data were not reliable enough to analyze. As reward learning may be a relevant construct to explore in youth with anxiety and sleep problems, future research should examine the role of reward learning in the maintenance and treatment of SRPs in children with anxiety using more reliable measures.

Conclusions and Future Directions

Despite the significant mental and physical health consequences of SRPs and the high prevalence of SRPs in school-aged children with anxiety (Alfano et al., 2007; Busch et al., 2017), little research has examined sleep treatments specifically tailored for this population and for this age group (Bourchtein et al., 2019; Busch et al., 2017; Meltzer, 2017). The current study demonstrates that it is possible to adapt sleep treatment to address the SRPs that are more common in children with anxiety by combining traditional sleep treatment elements (e.g., sleep hygiene) with traditional anxiety treatment elements (e.g., exposure, rewards). Future research is necessary to better understand and test the possible mechanisms (e.g., parent behaviors) maintaining SRPs in children with anxiety and how components of interventions adapted to this population might directly target these mechanisms.

Better understanding of the types of SRPs common in children with anxiety, mechanisms maintaining these SRPs, and the most effective ways to address them in treatment are important steps in addressing the mental and physical health consequences of SRPs (Alfano, 2018). Brief sleep treatments could be a cost-effective way to initiate change and improve symptoms for children with anxiety. Future research should investigate whether brief parent-focused sleep treatments such as this one also improve other child symptoms as well as more general family functioning and child physical health, and how they can be integrated into current treatments. For example, an in-person or online version of a brief, parent-focused sleep treatment may be appropriate while patients are on waiting lists for child anxiety treatment. Developing and disseminating these behavioral interventions for SRPs is imperative, as approximately 80% of children who present to outpatient health centers with SRPs are prescribed medication despite the fact that there are no FDA approved medications for children with insomnia (Badin, Haddad, & Shatkin, 2016; Vriend & Corkum, 2011). Thus, future research that expands and builds upon extant literature on SRPs in anxious youth is likely to have a significant positive impact on children's and families' overall health and wellbeing.

Tables

Table 1

Diagnostic Criteria Across Assessment Time Points

		Criteria A: Sleep Disturbance				Criteria B: Consequences of Sleep Disturbance				Criteria C: Adequate Opportunity	Criteria D: At least 3 Days a Week	Criteria E: At least 3 Months	Criteria F: Rule Out Other Disorders
Participant	Time Point	Met Criteria for Chronic Insomnia				B. Fatigue				C. Adequate opportunity/ circumstances	D. Days a week	E. Length of difficulty sleeping	F. Not explained by other sleep disorder
		A. Sleep Initiation	A. Sleep Dependence	A. Bedtime Resistance	A. Sleep Maintenance	B. Attention/ Concentration	B. Impaired functioning (e.g., school, social, family)	B. Irritability/sadness	B. Daytime sleepiness				
P1	T1	X	X	X		X		X	X	X	3	8 years	X
	T2				X		X	X		X	1.5	8 years	X
	T3	X	X		X		X	X	X	X	4	8 years	X
P2	T1	X	X	X		X	X	X		X	7	8 years	X
	T2	X	X	X						X	7	8 years	X

P3	T3			X									X	3.5	8 years	X
	T1	X	X	X		X		X	X	X	X	X	X	4	8 years	X
	T2				X								X	0.5	8 years	X
P4	T3				X	X	X			X			X	2.5	8 years	X
	T1	X	X	X	X	X		X	X	X	X		X	5	12 years	X
	T2				X			X		X			X	2	12 years	X
P5	T1	X	X					X	X	X	X	X	X	6	3 years	X
	T2		X					X		X		X	X	1.5	3 years	X
	T3		X							X			X	1	3 years	X
P6	T1	X	X				X	X	X	X	X	X	X	4.5	9 months	X
	T2		X					X	X		X	X	X	1	9 months	X
	T3		X				X			X			X	2	9 months	X
P7	T1	X	X					X	X	X	X	X	X	7	4 years	X
	T2		X							X	X		X	1.5	4 years	X
	T3		X										X	1.5	4 years	X
P8	T1	X	X		X			X	X	X	X		X	4	9 years	X
	T2		X							X	X		X	1.5	9 years	X
	T3	X	X		X				X	X			X	3	9 years	X
P9	T1	X		X				X	X		X	X	X	7	3 years	X
	T2												X	0	3 years	X
	T3												X	0	3 years	X
P10	T1	X		X						X		X	X	7	1 year	X
	T2		X							X			X	1.5	1 year	X
	T3	X			X			X		X	X	X	X	3.5	1 year	X

Note. T1=Baseline, T2=Post-Treatment, T3=Follow-up.

Table 2

Parent Qualitative Descriptions of Treatment Skills

Skill	Example Quote
Psychoeducation	“The image that really stuck with me...was the [habituation] curve of how it, you know, it gets to the top and [anxiety has to] come down and how we have to ease that curve down so that we don’t save her every single time...that image was really helpful to me even when she was having a bad night, that this will come down and it kept me calm...which in turn kept her calm.”
Sleep Hygiene	“[I learned] the importance of the same sleep time or bed time and even more so, the same wake time. That's like -- if only one could be absolutely fixed, you would fix wake time.”
Empathize and Encourage	“[I] learned to remain calm, and to use empathize and encourage, and that I don’t have to like try to fix it all for her exactly, and that I should not agree with her anxious statements...Not that I can't empathize with her, but that you know I shouldn’t be feeding into it.”
Positive Reinforcement	“[I learned how to use] the point system and kind of how to set that up and the difference between, using like toys or whatever, prizes, as a motivator versus just like bribery.”
Exposure	“[I learned how] to really break down the what the anxiety is ...so that you can actually practice small pieces rather than trying to practice the whole entire thing all at once.” Her daughter also describes breaking up the anxiety into smaller parts as in the example of practicing for going to a sleepover, “Were sort of switching around, so sometimes on the sofa, sometimes on the bed, like add different parts [to practice for the sleepover].”

Table 3

Parent-Child Sleep Interaction Scale (PSIS) Single Case Means and Effect Sizes

Participant		Sleep					
		Sleep Dependence		Reinforcement		Sleep Conflict	
		<i>M</i>	NAP	<i>M</i>	NAP	<i>M</i>	NAP
P1	Baseline	0.21		1.15		0.50	
	Treatment	0.26	0.45	0.80	0.41	0.63	0.45
	Post	0.00	0.57	0.20	0.25	0.40	0.56
	Follow-up	0.29	0.44	0.29	0.27	0.29	0.59
P2	Baseline	2.00		0.00		0.17	
	Treatment	1.54	0.67	1.34	0.81*	1.33	0.22* ^a
	Post	0.71	0.86*	0.71	0.64	0.71	0.42
	Follow-up	0.29	0.93*	2.71	1.00*	0.14	0.51
P3	Baseline	0.29		0.24		0.48	
	Treatment	0.00	0.64	1.22	0.78*	0.19	0.63
	Post	0.00	0.64	1.93	0.99*	0.07	0.68
	Follow-up	0.00	0.64	2.00	1.00*	0.00	0.71
P4	Baseline	0.71		0.00		0.10	
	Treatment	0.57	0.55	0.61	0.71*	0.14	0.48
	Post	1.00	0.44	2.00	1.00*	0.00	0.55
P5	Baseline	0.00		0.67		0.00	
	Treatment	0.19	0.40	1.62	0.74	0.00	0.50
	Post	1.00	0.00* ^a	3.86	1.00*	0.00	0.50
	Follow-up	0	0.50	4.00	1.00*	0.00	0.50
P6	Baseline	0.57		0.43		0.00	
	Treatment	0.38	0.52	0.46	0.49	0.21	0.40
	Post	0.21	0.59	0.93	0.65	0.07	0.46
	Follow-up	0.57	0.43	1.00	0.79*	0.57	0.21* ^a
P7	Baseline	0.62		0.95		0.00	
	Treatment	0.00	0.69*	1.00	0.52	0.00	0.50
	Post	0.00	0.69	0.83	0.46	0.00	0.50
	Follow-up	0.14	0.63	0.14	0.46	0.00	0.50
P8	Baseline	0.36		0.43		0.07	
	Treatment	0.14	0.55	0.00	0.39	0.05	0.51
	Post	0.57	0.41	0.43	0.48	0.57	0.39
	Follow-up	0.00	0.61	0.00	0.39	0.00	0.54
P9	Baseline	0.55		0.82		0.00	

P10	Treatment	0.68	0.40	1.32	0.66	0.24	0.42
	Post	0.43	0.49	1.86	0.79*	0.00	0.50
	Follow-up	0.00	0.68	1.71	0.75	0.00	0.50
	Baseline	0.43		2.00		1.00	
	Treatment	0.13	0.59	2.87	0.75	0.80	0.51
	Post	0.00	0.64	2.42	0.57	0.08	0.76
	Follow-up	0.00	0.64	1.29	0.26	0.29	0.68

Note. * $p < .05$. NAP for treatment means all baseline points compared to all treatment points, ^a represents a significant deterioration rather than improvement (e.g., increase in sleep dependence or sleep conflict).

Table 4

Parenting Outcomes at Baseline, Post-Treatment, and Follow-up

Measure	Baseline (N=10)	Post (N=10)	Z	r	Baseline (N=9)	Follow-up (N=9)	Z	r
	Mean (SD)	Mean (SD)			Mean (SD)	Mean (SD)		
Total Accom	3.40 (2.27)	1.90 (1.52)	-2.56*	0.81	3.67 (2.24)	2.00 (1.73)	-2.15*	0.72
Total Accom Interference	9.30 (6.67)	6.20 (6.14)	-1.68	0.53	10.22 (6.36)	5.67 (6.61)	-2.03*	0.68
Parenting Scale Total Score	3.08 (0.40)	2.71 (0.43)	-2.50*	0.79	3.11 (0.41)	2.63 (0.64)	-2.07*	0.69
Parenting Scale Laxness Score	2.75 (0.69)	2.54 (0.45)	-1.38	0.44	2.74 (0.73)	2.42 (0.71)	1.90	0.63
Parenting Scale Overreactivity Score	2.92 (0.80)	2.41 (0.60)	-1.61	0.51	3.00 (0.80)	2.58 (0.78)	-1.54	0.51
Parenting Scale Verbosity Score	3.67 (0.76)	3.34 (0.80)	-1.48	0.47	3.74 (0.76)	3.06 (1.08)	-1.84	0.61

Note. *= $<.05$.

Table 5

Sleep, Anxiety, and General Functioning Outcomes at Baseline, Post-Treatment, and Follow-up

	Measure	Baseline (N=10)	Post (N=10)	Z	r	Baseline (N=9)	Follow-up (N=9)	Z	r
		Mean (SD)	Mean (SD)			Mean (SD)	Mean (SD)		
∞	Clinical Severity Rating	5.15 (0.63)	2.45 (0.93)	-2.84*	0.90	5.06 (0.58)	2.89 (1.62)	-2.53*	0.84
	Total CSHQ Score	53.80 (5.94)	44.30 (7.45)	-2.81*	0.89	53.44 (6.19)	44.89 (6.11)	-2.55*	0.85
	CSHQ Sleep Anxiety	8.20 (1.48)	6.30 (2.11)	-2.54*	0.80	8.22 (1.56)	6.11 (1.76)	-2.34*	0.78
	CSHQ Bedtime Resistance	9.30 (2.26)	7.30 (2.45)	-2.71*	0.86	9.33 (2.40)	7.89 (2.15)	-2.21*	0.74
	CSHQ Sleep Onset Latency	2.60 (0.52)	1.70 (0.82)	-2.46*	0.78	2.67 (0.50)	1.78 (0.83)	-2.53*	0.84
	CSHQ Sleep Duration	6.30 (1.34)	4.10 (1.52)	-2.58*	0.82	6.22 (1.39)	4.22 (1.20)	-2.45*	0.82
	CSHQ Night Wakings	5.00 (1.63)	4.10 (0.99)	-1.28	0.40	5.00 (1.73)	3.33 (0.50)	-2.21*	0.74
	CSHQ Daytime Sleepiness	13.90 (3.00)	11.90 (2.69)	-2.51*	0.79	13.44 (2.79)	12.56 (2.60)	-1.28	0.43
	CSHQ Parasomnias	9.20 (1.32)	8.60 (0.84)	-1.29	0.41	9.22 (1.39)	8.67 (1.00)	-1.52	0.51
	CRSP Insomnia	13.40 (3.60)	11.90 (3.07)	-1.77	0.56	13.78 (8.00)	13.00 (3.57)	-0.51	0.17
	CRSP Bedtime Worries	5.20 (2.62)	4.00 (2.79)	-1.51	0.48	5.44 (2.65)	5.22 (3.11)	-0.21	0.07
	CRSP Sleep Location	9.40 (3.06)	8.30 (3.30)	-1.48	0.47	9.44 (3.25)	8.00 (2.83)	-1.71	0.57
	CRSP Bedtime Activities	12.70 (3.83)	14.00 (4.42)	-1.49	0.47	12.00 (3.32)	12.67 (2.29)	-0.78	0.26
	CRSP Caffeine	5.00 (2.21)	4.20 (1.48)	-1.81	0.57	4.89 (2.31)	4.00 (1.58)	-1.63	0.54
	CRSP Electronics	4.70 (2.36)	4.50 (2.59)	-0.71	0.22	4.67 (2.50)	4.00 (1.41)	-0.81	0.27

CRSP Restless Legs	9.60 (2.12)	10.80 (2.49)	-1.19	0.38	9.89 (2.03)	9.78 (2.64)	-0.43	0.14
CRSP Parasomnias	5.30 (1.06)	5.40 (0.97)	-0.45	0.14	5.56 (0.73)	5.33 (1.00)	-1.00	0.33
Parent MASC Total Score	59.90 (17.78)	51.50 (16.12)	-1.79	0.57	60.67 (18.68)	48.78 (16.13)	-2.20*	0.73
Parent MASC Separation Anxiety	14.70 (3.53)	10.80 (4.44)	-2.19*	0.69	14.44 (3.64)	10.44 (4.75)	-2.32*	0.77
Child MASC Total Score	50.90 (19.34)	45.40 (19.66)	-1.67	0.53	53.56 (18.47)	47.33 (20.74)	-1.19	0.40
Child MASC Separation Anxiety	11.20 (5.43)	10.30 (4.97)	-0.77	0.24	12.00 (5.10)	11.33 (6.98)	-1.02	0.34
Pediatric Symptom Checklist	21.11 (6.45) ^a	15.33 (11.12) ^a	-2.10*	0.70	20.67 (5.61)	17.22 (7.33)	-2.04*	0.68
Brief Impairment Total	10.83 (6.26)	9.76 (6.41)	-1.12	0.35	9.70 (5.45)	8.26 (5.51)	-1.01	0.34

Note. * $p < .05$. ^a $n = 9$ for the Post-Treatment Pediatric Symptom Checklist.

Table 6

Sleep Onset Latency Single Case Means and Effect Sizes

Participant		Actigraphy		Sleep Diary	
		<i>M</i>	NAP	<i>M</i>	NAP
P1	Baseline	10.46		34.73	
	Treatment	15.10	0.34	14.33	0.77*
	Post	14.33	0.39	17.33	0.71
	Follow-up			23.57	0.63
P2	Baseline	66.17		61.67	
	Treatment	48.29	0.68	56.19	0.51
	Post	45.80	0.63	66.00	0.43
	Follow-up			53.57	0.54
P3	Baseline	25.74		49.15	
	Treatment	16.33	0.65	24.44	0.78*
	Post	8.09	0.77*	18.62	0.88*
	Follow-up	1.43	0.90*	15.00	0.93*
P4	Baseline			30.11	
	Treatment			25.88	0.45
	Post			16.67	0.55
	Follow-up				
P5	Baseline	38.00		52.50	
	Treatment	12.81	0.95*	33.86	0.83*
	Post	1.71	1.00*	25.00	0.94*
	Follow-up			20.71	0.98*
P6	Baseline	22.40		51.00	
	Treatment	24.75	0.50	38.27	0.74*
	Post	18.92	0.62	22.5	0.96*
	Follow-up			68.57	0.41
P7	Baseline	24.65		48.81	
	Treatment	14.45	0.73*	44.44	0.55
	Post	10.83	0.83*	35.83	0.69
	Follow-up			30.71	0.86*
P8	Baseline	12.83		19.23	
	Treatment	15.71	0.40	11.76	0.54
	Post	15.86	0.41	24.29	0.69
	Follow-up			13.00	0.68
P9	Baseline			46.00	

P10	Treatment			53.33	0.45
	Post			18.57	0.91*
	Follow-up			27.14	0.81*
	Baseline	45.17		63.33	
	Treatment	34.64	0.58	32.00	0.80*
	Post	17.29	0.79*	23.64	0.83*
	Follow-up			61.71	0.50

Note. * $p < .05$. NAP score for treatment means all baseline points compared to all treatment points.

Figures

Figure 1

Assessment Flowchart

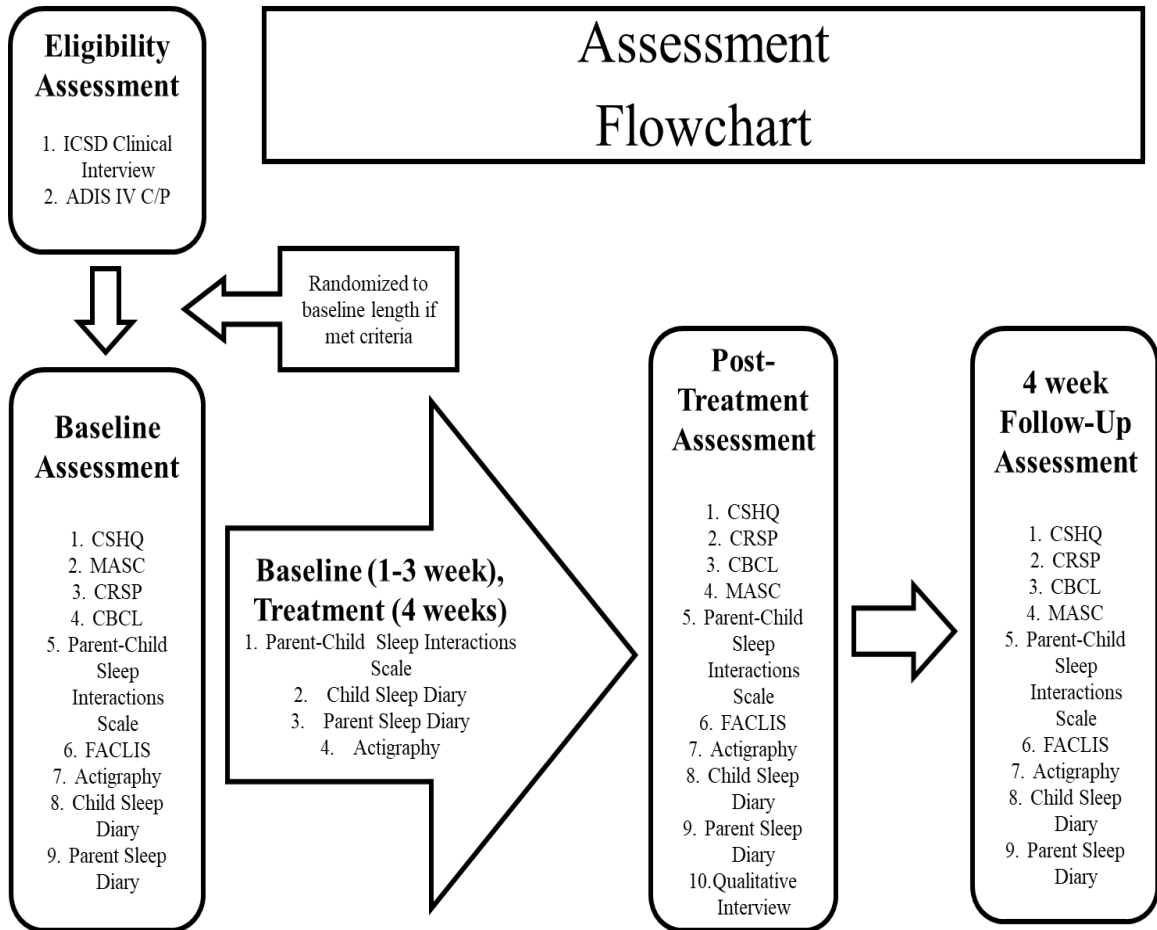


Figure 2

Sleep Reinforcement Single Case Graphs

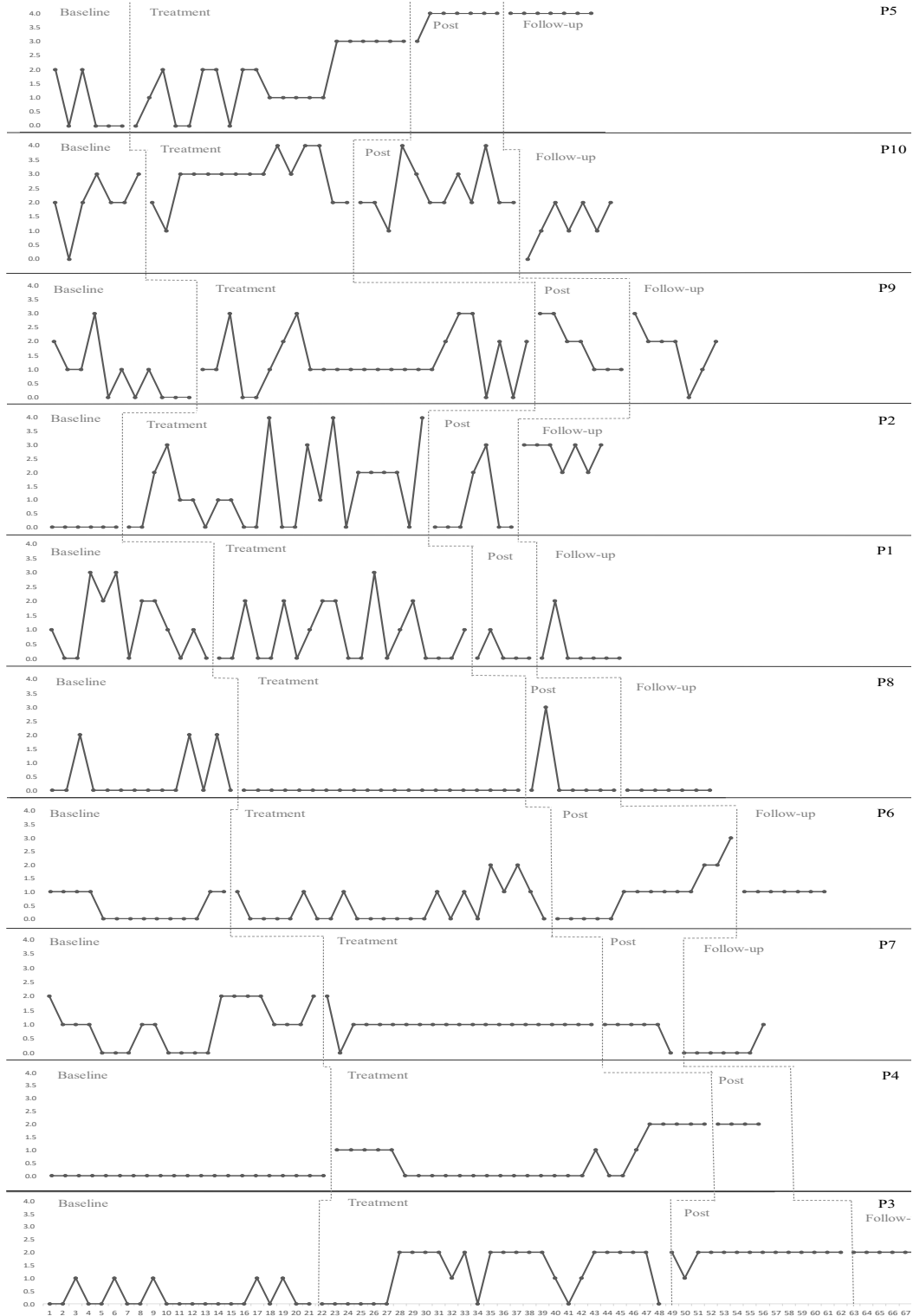
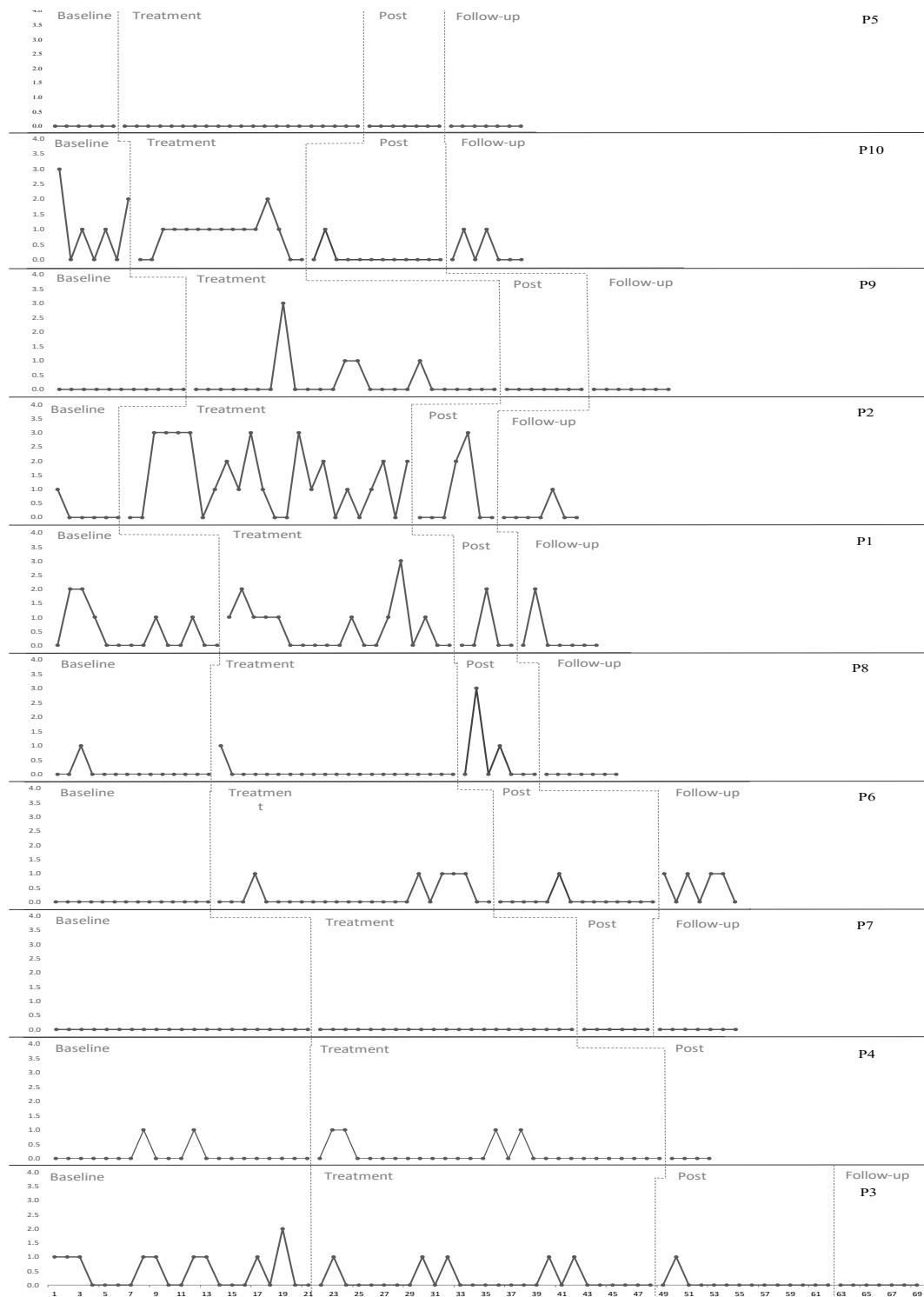


Figure 3

Sleep Conflict Single Case Graphs



Sleep Dependence Single Case Graphs

Sleep Dependence Single Case Graphs

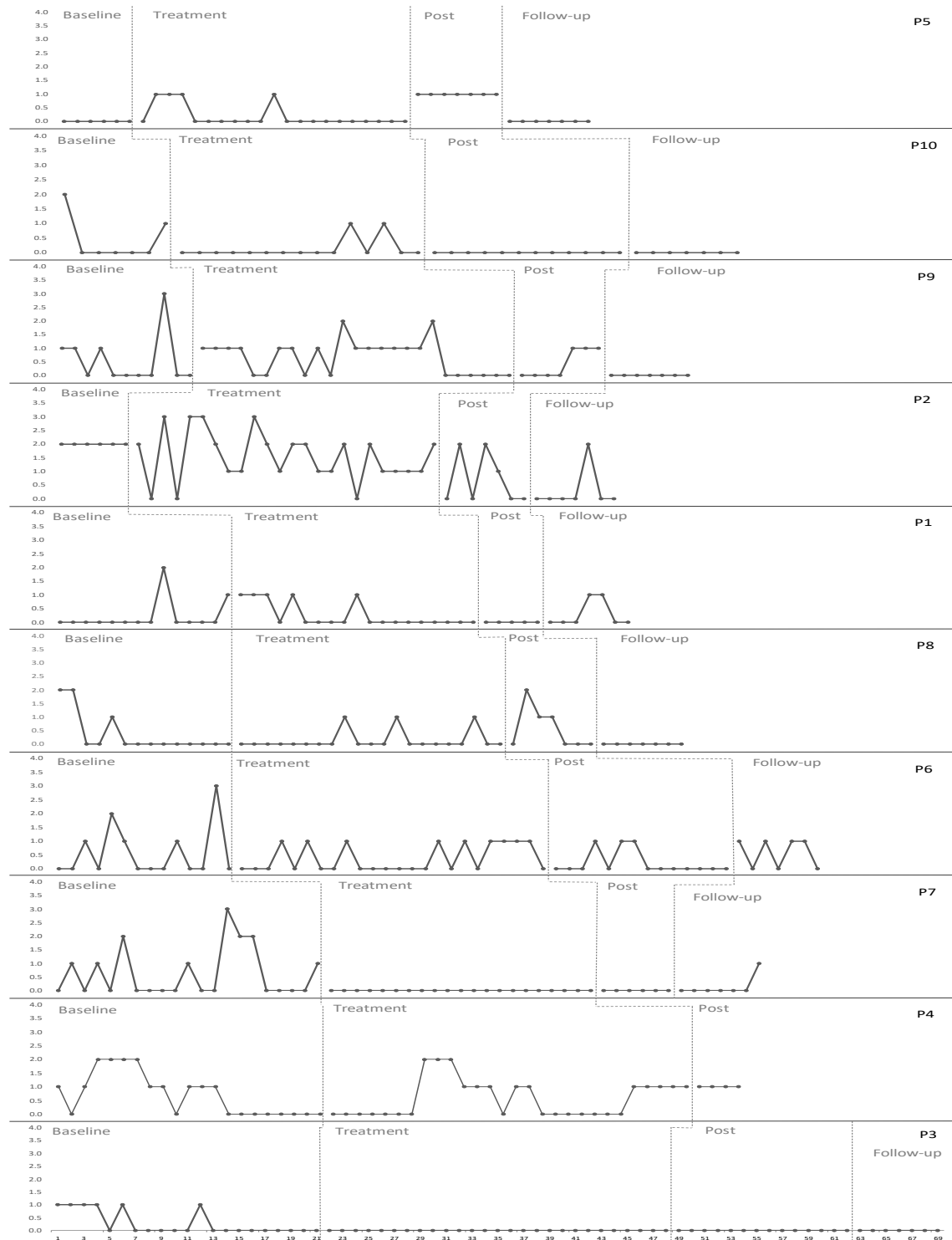


Figure 5

Subjective Sleep Onset Latency Single Case Graphs

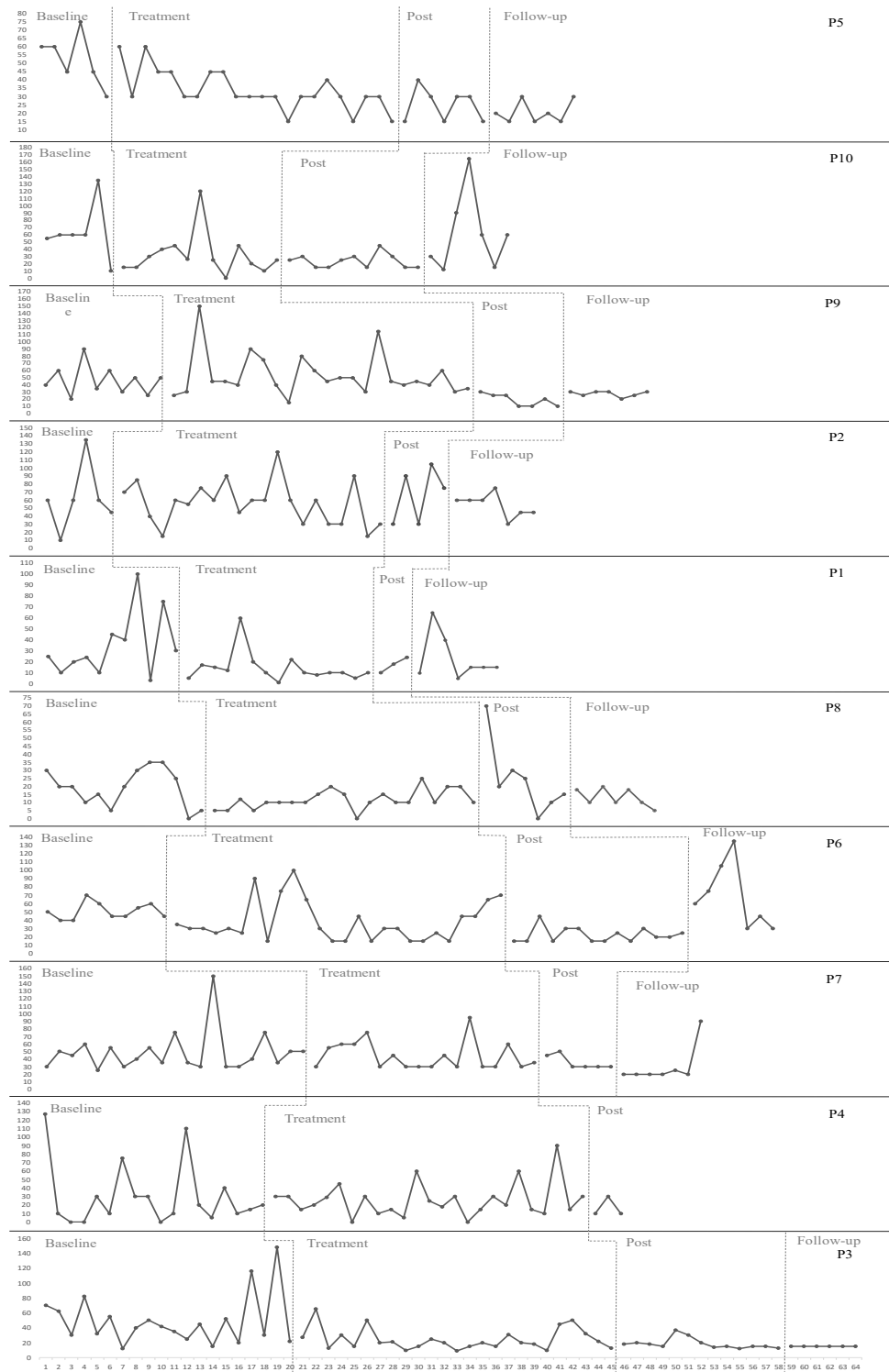
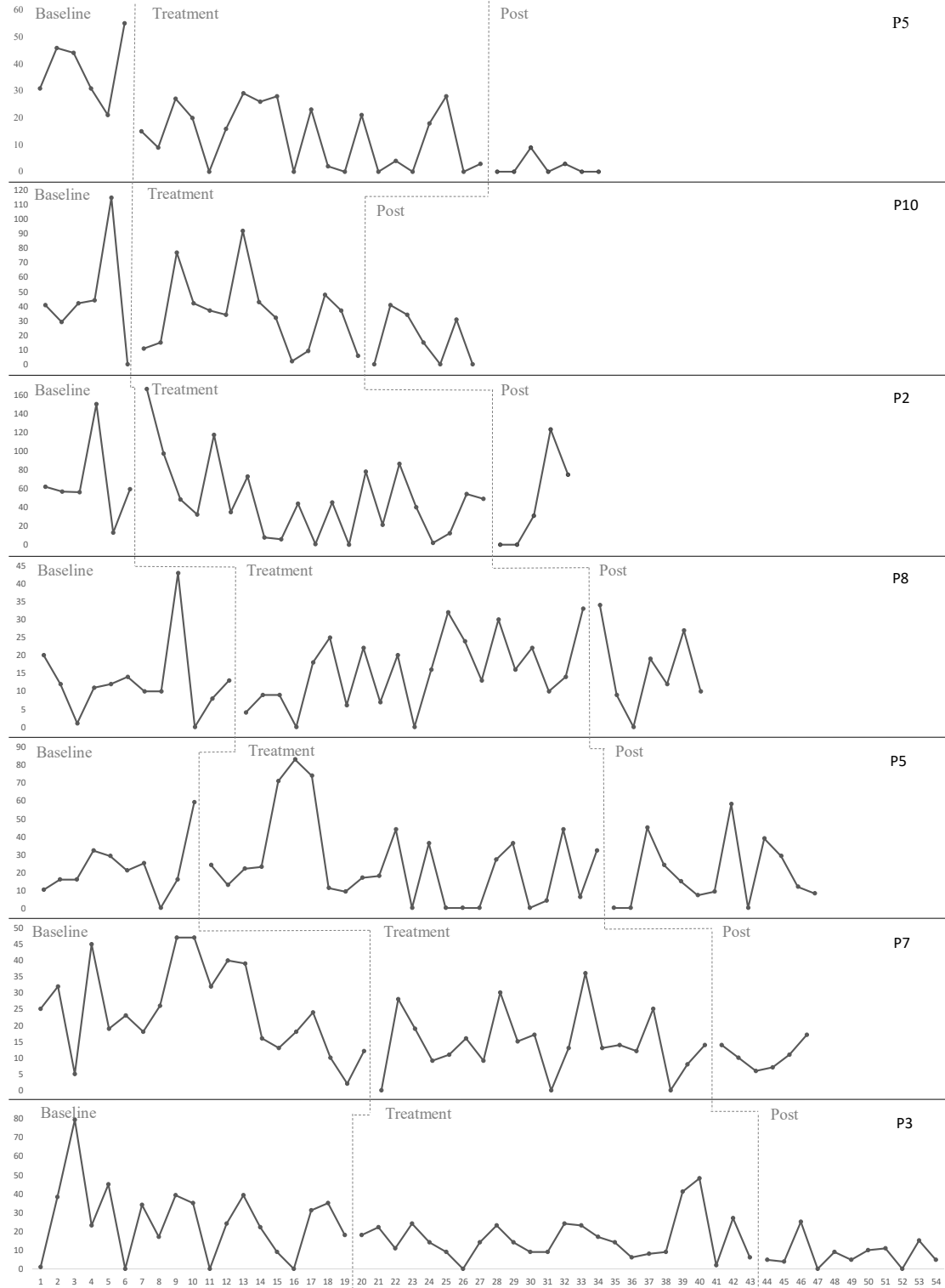


Figure 6

Objective Sleep Onset Latency Single Case Graphs



Appendices

Appendix A: Phone Screens

Brief Phone Screen (completed as part of normal Child Center for Anxiety phone screen)

1. How well has your child been sleeping recently?
2. Is it ever hard for your child to fall asleep at night?
3. How long does it usually take your child to fall asleep at night?
4. Does your child ever wake up in the middle of the night?
5. Does your child need parental assistance to fall asleep?
6. Does your child worry about sleep (e.g., getting enough, being able to fall asleep)?
7. Would you be interested in treatment related to these concerns?

Detailed Phone Screen

Hello, my name is Lydia Chevalier and I am one of the graduate students at the Center for Anxiety and Related Disorders. I am conducting a study to understand the types of sleep difficulties that children with anxiety have and how best to treat them. If you are interested in being considered for the study, I can ask you some brief questions about your child's sleep. If it seems like this study would be a good fit, I can then explain what the study would entail to see if you and your child would be interested in joining.

1. How well has your child been sleeping recently?
2. Is it ever hard for your child to fall asleep at night?
3. How long does it usually take your child to fall asleep at night?
4. Does your child require parent intervention to fall or stay asleep during the night?
5. Does your child ever wake up in the middle of the night?
6. What time does your child usually get up in the morning?
7. How energetic does your child seem when he/she wakes up in the morning? (e.g., tired, energized)
8. When your child hasn't slept well do you ever notice whether they:
 - a. Feel sleepy during the day?
 - b. Get in trouble at school?
 - c. Feel grumpy or irritable?
 - d. Feel sad or down?
 - e. Have trouble sitting still?
 - f. Get angry easily?

- g. Have trouble remembering things?
- 9. Does your child worry about not being able to sleep?
- 10. How long has your child been having trouble sleeping? (3 or more months)
- 11. How many times a week does one of these problems with sleep occur? (3x a week or more)
- 12. Have you ever noticed your child snoring during the night?
 - a. If yes, how often?
 - b. If yes, does this snoring occur year round or only when child has allergies or a cold?
- 13. Does your child regularly wet the bed during the night? a. If so, how often?
- 14. Have you received parent training to help with your child's sleep in the last year or has your child had any treatment for sleep in the last year?
- 15. Is your child currently engaged in treatment for sleep or anxiety?
- 16. Has your child ever been diagnosed with ADHD or depression?

If a good fit: Thanks so much for providing that information, it looks like your family would be a good fit for this study. If you are enrolled in the study, we will assign you by chance (like a coin toss) to tracking sleep using the sleep diaries and actigraphy watches for 1 week, 2 weeks, or 3 weeks before the study treatment begins. You and the researcher cannot choose the length of tracking before treatment; you will have an equal chance of being placed in each time length. After this initial period of tracking sleep patterns, you will attend four 90-minute parent-training sessions within an 8-week time period. These treatment sessions will teach you to help your child develop healthier sleeping habits. It will include skills such as developing consistent bedtimes and bedtime routines, reducing reliance on parental presence when falling asleep, reducing co-sleeping, and using rewards to encourage behaviors like staying in bed throughout the night. You will also receive one 30-minute phone check-in each week to help problem solve around using these skills. These parent-training sessions and phone check-ins will be provided to you as part of the study and at no cost to you. During this treatment period you will fill out one questionnaire (12 questions), complete a sleep diary with your child about his/her sleep, and answer 4 questions about your sleep, which should collectively take approximately 15 minutes a day. You will also be asked to time bedtime routine each night and wear an actigraphy watch so that we can better understand the relationship between your sleep and your child's sleep. Your child will also wear an actigraphy watch during this treatment phase but will not have to come in to the Center. Does this sound like something that you are still interested in? If yes-set up the first assessment.

If not a good fit: Thanks so much for providing that information, it looks like this study may not be the best fit for you because XXXX. You are still on the waitlist for the anxiety treatment that you originally called for and the clinician you will be paired with will be able to help you address many of these concerns. Thanks so much for your time and feel free to contact us at 617-353-9610 if you have any questions.

**Appendix B: Chronic Insomnia Interviews Developed for this Study to Assess for
Chronic Insomnia Based on the International Classification of Sleep Disorders
(ICSD) Criteria**

Parent ICSD Chronic Insomnia Interview

1. How has your child's sleep been recently?
2. Where does your child usually sleep? (e.g., own bed, parent's bed)
3. What time does your child usually start getting ready for bed?
4. What does your child do to get ready for bed? (e.g., brush teeth, read a book)
5. What time does your child usually get into bed?
6. Does your child resist going to bed at his/her bedtime? (e.g., beg to stay up later, start crying/tantruming, repeatedly reassurance seek). If yes, how so?
7. Is your child unable to fall asleep without you or another caregiver present? (e.g., parent stayed until child fell asleep, child slept on bed in parent's room, co-slept with parent) If yes, follow up on basic details.
8. How long does it usually take your child to fall asleep at night?
9. Is it ever hard for your child to fall asleep at night? What makes it hard for your child to fall asleep? (e.g., too much noise, scared)
10. Does your child ever wake up in the middle of the night?
11. If so, what do you do when your child wakes up (e.g., allow them to sleep in parent's bed)
12. What time does your child usually get up in the morning?
13. What does your child do when he/she first wakes up? (e.g., lay in bed, get up and get dressed)
14. How energetic does your child seem when he/she wakes up in the morning? (e.g., tired, energized)
15. Approximately how many nights a week does your child have trouble falling or staying asleep, resist bedtime, or need a parent present to fall asleep? (3x or more a week)
16. Of those (#) nights how many of the next days did you notice that your child:
clarify that these must be more on days they didn't sleep than on other days
 - a. Feels sleepy during the day?
 - b. Gets in trouble at school?
 - c. Feels grumpy or irritable or gets angry easily?
 - d. Feels sad or down?
 - e. Has trouble sitting still?

- f. Has trouble remembering things or concentrating?
 - g. Has trouble functioning at school?
 - h. Is more prone to errors/accidents?
 - i. Is anxious about sleep (e.g., not getting enough sleep, consequences of sleep disturbance)?
17. On a scale of 0 to 8, how much would you say the sleep problems have interfered with your child's life? That is, how much they have interfered with your child's friendships, caused problems at school or home, or stopped your child from doing the things he/she would like to do.
18. What have his/her sleep problems interfered with?
19. How long has your child been having trouble sleeping? (3 or more months)
20. Have you ever noticed your child snoring during the night?
- a. If yes, how often?
 - b. If yes, does this snoring occur year round or only when child has allergies or a cold?
21. Does your child regularly wet the bed during the night? a. If so, how often?
22. Does your child use Melatonin or any other medication that affects sleep? (e.g., medications for ADHD, Benadryl) a. If so, how often?
23. Have you received parent training to help with your child's sleep in the last year or has your child had any treatment for sleep in the last year?
24. Is your child currently engaged in treatment for sleep or anxiety?

Child ICSD Chronic Insomnia Interview

1. How has your sleep been recently?
2. Where do you usually sleep (e.g., own bed, parent's bed)
3. What time do you usually start getting ready for bed?
4. What do you do to get ready for bed? (e.g., brush your teeth, read a book)
5. What time do you usually get into bed?
6. How long does it usually take you to fall asleep at night?
7. Do you ever need your parent with you when you fall asleep at night?
8. Is it ever hard to fall asleep at night?
9. What makes it hard to fall asleep? (e.g., too much noise, scared)
10. Do you ever wake up in the middle of the night?
11. If so, what do you do when you wake up (e.g., go back to sleep, go to parent's room)
12. What time do you usually get up in the morning?
13. What do you do when you first wake up? (e.g., lay in bed, get up and get dressed)
14. How do you feel when you wake up in the morning? (e.g., tired, energized)
15. About how many nights a week do you have trouble falling or staying asleep?
16. Of those (#) nights how many of the next days do you notice that you: ****clarify that these must be more on days they didn't sleep than on other days****
 - a. Feel sleepy during the day?

- b. Get in trouble at school or at home?
 - c. Feel grumpy or irritable or get angry easily?
 - d. Feel sad or down?
 - e. Have trouble sitting still?
 - f. Have trouble remembering things or concentrating?
 - g. Have trouble functioning at school?
 - h. Made lots of mistakes?
 - i. Were worried about sleep (e.g., not getting enough sleep, what will happen if you can't fall asleep)?
17. On a scale of 0 to 8, how much would you say the sleep problems have interfered with your life? That is, how much it has interfered with your friendships, caused problems at school or home, or stopped you from doing the things you would like to do.
18. What have your sleep problems interfered with?
19. How long have you been having trouble sleeping?

Appendix C: Detailed Treatment Session Guides

Session 1: Psychoeducation and Sleep Hygiene

1. Review findings from assessment
 - a. Diagnosed with:
 - a. Diagnoses are technical terms that allows doctors and mental health professionals can quickly and clearly communicate with each other about exactly what is going on with your child
 - b. Having a diagnosis helps us to recommend the best treatments for that particular set of behaviors and symptoms
 - c. Does not mean that diagnoses are permanent, or indicators that there is “something wrong” with your child, just that this is a current area of difficulty
2. Discuss the plan for this brief treatment
 - a. In these four sessions we will be focusing on your child’s sleep difficulties and teaching you strategies that you can use at home with your child to help them develop healthy sleep habits
 - b. As a reminder, we will meet four times for 1.5 hours to learn these skills and you and child will wear the watches and fill out questionnaires during this time
 - c. Ideally these four sessions will be weekly four in a row but we know that this is not always possible, they must all be within 8 weeks, and ideally within 6 weeks
 - d. At the end of the four sessions you and your child will come in for another assessment (will take only about 2 hours this time)
 - e. Four weeks after this assessment you and your child will come in for one more assessment (will take only about 1 hour this time) and wear the watches and do questionnaires for one more week
 - f. After that if you are interested in getting treatment at the Center (at regular cost) we can set you up with the next clinician that is available
 - g. Any questions?
3. Check in on how the actigraphy monitors, sleep diaries, and questionnaire went
 - a. How was wearing the actigraphy monitor?
 - i. Any issues?
 - b. How did your child do with wearing the actigraphy monitor?
 - i. Any issues?
 - c. How did filling out the child sleep diary go?
 - i. Any issues with remembering to complete it?

- d. How did filling out your sleep diary go?
 - i. Any issues with remembering to complete it?
 - e. How did filling out the Parent-Child Sleep Interactions Scale go?
 - i. Any issues with filling it out?
- 4. Psychoeducation on:
 - a. Normative sleep
 - i. *Handout:* Sleep in School-Aged Children (Mindell & Owens, 2003)
 - b. Interaction between sleep and anxiety
 - i. *Handout:* Anxiety is Like a False Fire Alarm (Parenting Group)
 - ii. *Handout:* Where do parents fit in? (Parenting Group)
 - c. Accommodation and effects on sleep
 - i. *Handout:* Parenting traps (Parenting Group)
 - ii. *Handout:* Taking Charge of Your Child's Sleep (University of Kansas)
- 5. Sleep Hygiene
 - a. *Handout:* Sleep Hygiene for Children (Children's Hospital of Orange County)
 - b. Establish regular bedtime routine
 - i. *Handout:* Taking Charge of Your Child's Sleep (University of Kansas)
 - c. Establish regular bedtime and wake-time
 - i. *Handout:* Taking Charge of Your Child's Sleep (University of Kansas)
 - ii. Faded bedtime
 - 1. Bedtime set to when child likely to fall asleep with 15 minutes and pushed back earlier and earlier until desired bedtime is reached
 - a. If can't fall asleep within 20 minutes removed to do non-activating/non-rewarding activity
 - iii. *Handout:* Bedtime Fading (APA)
- 6. Homework: Select one item of accommodation/sleep hygiene to address (e.g., establish regular bedtime and wake time) and complete questionnaire on handouts
 - a. Homework assigned:

Session 2: Reinforce Appropriate Behaviors

- a. Review homework from session 1 and problem solve
- b. Reinforcing behaviors
 - i. Empathize and encourage
 - 1. *Handout*: Empathize and Encourage (Parenting Group)
 - 2. *Handout*: Empathize and Encourage Practice at Home (Parenting Group)
 - ii. Effective Praise
 - 1. *Handout*: Tools for Effective Praise (Parenting Group)
- c. Introduce rewards
 - iii. Rewards for:
 - 1. Following bedtime routine
 - 2. Limited reassurance seeking
 - 3. Staying in bed
 - 4. Sleeping in bed alone/sleeping in own bed
 - 5. Sleeping without light on
 - 6. Getting out of bed on time
 - 7. Not napping
 - iv. *Handout*: Defining Successful Rewards (Parenting Group)
 - v. *Handout*: Sample Rewards (Parenting Group)
 - vi. *Handout*: Sample Reward Chart (Parenting Group)
- d. Briefly introduce the concept of exposures and plan out one exposure
 - 1. *Handout*: Habituation Curve
- e. Homework: Select one item of accommodation/avoidance to address (e.g., use reward system to reinforce staying in bed throughout the night) and complete questionnaire on handouts
 - a. Homework:

Session 3: Reduce Accommodation

- a. Review homework from session 2 and problem solve
- b. Reduce accommodation
 - vii. Limit Setting
 1. Limit how late child can watch TV/use electronics
 2. Limit how many times child can ask reassurance seeking questions
 3. Limit daytime napping
 4. Limit play/electronics use in bed during the day/evening
 5. Limit caffeine (e.g., chocolate, soda) intake before bedtime
 6. Limit accidental rewards for not sleeping (e.g., stay home from school, go in to school late)
 7. *Handout: Sleep in School-Aged Children* (Mindell & Owens, 2003)
 - viii. Exposure
 1. Gradually fade parent out of sleeping with child, child out of sleeping with parent, child sleeping with the light on etc.
 2. *Handout: Sample Fear Hierarchy*
 3. *Handout: Fear and Avoidance Hierarchy with Rewards*
 4. Pass System
 - a. Child has 3 “passes” that they can use to get a drink, hug, go to the bathroom after which parent no longer engages in accommodation
 - b. *Handout: Taking Charge of Your Child’s Sleep* (University of Kansas)
- c. Homework: Select one item of accommodation to address (e.g., use pass system to reduce reassurance seeking questions) and complete questionnaire on handouts
 - a. Homework:

Session 4: Review and Maintenance/Relapse Prevention

- a. Review skills and problem solve
 - ix. Sleep hygiene
 - 1. Regular bedtimes and wake times
 - 2. Bedtime fading
 - x. Reduce accommodation
 - 1. Limit setting
 - 2. Exposure
 - xi. Encourage appropriate behaviors
 - 1. Empathize and encourage
 - 2. Effective praise
 - 3. Rewards
- b. Maintenance/Relapse Prevention
 - xii. Plan for continued rewards, limits, and exposures
 - xiii. Plan for changes in schedule/location (e.g., vacations, school starting/ending)
 - xiv. *Handout: Getting on the Same Page* (Parenting Group)
- c. Discuss plan going forward
 - a. Next you and your child will come in for another assessment (will take only about 2 hours this time)
 - b. Four weeks after this assessment you and your child will come in for one more assessment (will take only about 1 hour this time) and wear the watches and do questionnaires for one more week
 - c. After that if you are interested in getting treatment at the Center (at regular cost) we can set you up with the next clinician that is available

Appendix D: Post-Treatment Qualitative Interview- Parent

Goals & Purpose: The interviewers will conduct semi-structured qualitative interviews to elicit parents' feedback about the assessment and treatment during the study as well as suggestions for improving both. Broad-based probes will focus on the clinical interviews about anxiety and sleep, the computer task to measure reward learning, sleep diaries, actigraphy, questionnaires, treatment components, changes made due to treatment, importance of these changes, confidence in using skills learned in treatment, and suggestions for improving treatment for other families. Questions will be open-ended when possible to encourage informants to elaborate on issues they consider important.

Data Collection & Preliminary Analysis: Interviews will be conducted by study staff in a private treatment room at the Center for Anxiety and Related Disorders. Interviews will be completed by clinical doctoral students or highly trained undergraduate research assistants, average 1 hour, and will be audio recorded or video recorded and transcribed.

Data analysis will consist of applied thematic and content analysis. Immediately after the data collection appointment (within 24 hours or before the next interview, whichever is sooner), the interviewer will review her handwritten notes taken during the interview and briefly summarize the interview dynamics and possible themes for further study. Once the interviews are transcribed, the person coding the transcripts will expand the summary based on content and thematic analysis. Participants will be identified by their study ID number in reports and transcripts. The interview guide and intent statements will serve as the basic structure for organizing the final report. Key points will be summarized on a question-by-question basis.

Equipment & Materials: The interviewer will take two digital recorders, extra batteries, and hard copy of the interview schedule (for note-taking). The interviewer will also schedule the video recording for the clinical room selected for this interview.

Intent Statements: Each section of this guide is preceded by an intent statement which explains the type of data that the section has been designed to gather. Intent statements are designed so that all interviewers focus their discussion on the same areas of interest and thus gather similar data. This will allow the interviewer to deviate from the script for a more naturally flowing conversation while also ensuring that relevant themes are being captured.

Interview Questions: There are two primary types of questions noted in the agenda: open-ended, "lead" questions and probes which tend to be less open-ended. Both lead questions and probes are intended to guide the interviewer to gather all the data investigators are seeking. They are not meant to be used verbatim. Interviewers should consider the best way to gather the data and rephrase questions to address intents and local cultural norms and practices.

Interview Schedule

- I. Preliminary Information & Introductions:** *The intent of this portion of the agenda is to welcome the parent and make him/her as comfortable as possible by discussing what to expect during the interview and who hears the data (just the research team). Explain that the data will be transcribed and analyzed anonymously (e.g., “we will write down everything you say, but we will not write down your name – your data will be connected to a unique identifier”). Explain that their child’s future care at the Center will not be affected by anything that they say during this interview.*

A. Introductions & Purpose of Interview:

1. Thank you for meeting with me to discuss your thoughts about treatment. As you know, one of our goals is to learn about how best to help parents help their children develop healthy sleep habits. This will be a short meeting so that I can hear your impressions. Your feedback is important to us and will help us improve how we work with families, particularly those of children with sleep and anxiety concerns. Please feel free to share openly. There are no right or wrong answers and your responses will not impact your ability to access care at our Center going forward.
- First I’m going to ask you about the assessment, when you came into our Center the first time to answer questions about your child and his/her sleep. Next I will ask you about times you met with Lydia and what was helpful or not as helpful. Last, I will ask you if you have any suggestions for how we could improve the treatment.

B. Housekeeping:

1. You can take a break or ask to turn off the recorder at any time.
2. Please tell me if you feel uncomfortable for any reason.
3. You can stop the interview at any time – you do not have to keep going if you choose not to.
4. If I ask you something that doesn’t really make sense, please tell me, and I can ask the question in another way.

C. Audio recordings:

1. Your thoughts and experiences are very important to me. Therefore, I would like to record our conversation. This is so that I remember what you say later on.
2. I am going to turn on the audio recorders. Try to just ignore the recorders.

II. **Assessment:** *The intent of this portion of the interview is obtain an understanding of the parent's and child's experience during the assessment. The purpose of gathering this information is to understand how to improve assessments and also get a sense of the parent's understanding of why each assessment component was used. Examples of details to gather for each assessment component are initial thoughts about the assessment component, any concerns about the assessment component, any way to improve aspects of the assessment or compliance with instructions, and ideas about why this assessment component was conducted.*

A. Lead Question: You and your child came to our Center for an initial assessment about a month ago and completed another assessment today. What are your general thoughts about those assessments?

1. Follow-Up Question 2: You and your child were asked to fill out a sleep diary together for every night of the study in order to track your child's sleep. What did you think about this sleep diary?

a. Probes:

1. What do you think the purpose of the sleep diary was in the assessment?
 - I. During treatment?
2. What, if anything, did you learn from completing a sleep diary with your child?
3. Did anything surprise you?
 - I. If yes, what?
4. On a scale of 1 (never) to 10 (always) how often were you able to remember to fill out the sleep diary with your child?
5. What barriers were there to completing the sleep diary each day?

2. Follow-Up Question 3: You were asked to complete your own brief sleep diary in order to track your sleep in relation to your child's. What did you think about completing a sleep diary for yourself?

a. Probes:

1. What, if anything, did you learn from completing your own sleep diary?
2. Was anything surprising?
 - I. If yes, what?
3. Did you make any changes to your sleep based on things you learned from your sleep diary?
 - I. If yes, what changes did you make?
 - II. Did these changes impact your sleep?

4. On a scale of 1 (never) to 10 (always) how often were you able to remember to fill out your sleep diary every day?
5. What barriers were there to completing your sleep diary each day?
3. Follow-Up Question 4: During the study your child was asked to wear an actigraphy watch to track his/her sleep. What did you think about your child wearing this watch?
 - a. Probes:
 1. What do you think the purpose of your child wearing the actigraphy watch was?
 2. What barriers were there to your child wearing the actigraphy watch (e.g., felt that it was uncomfortable, concerned your child would lose it)?
4. Follow-Up Question 5: During the study you were asked to wear an actigraphy watch. What did you think of wearing this watch?
 - a. Probes:
 1. What do you think the purpose of you wearing the actigraphy watch was?
 2. What barriers were there to wearing the actigraphy watch (e.g., felt that it was uncomfortable, did not look professional)?

III. **Treatment Components:** *The intent of this section is to get feedback on the parent training treatment that was conducted. We want to hear about what skills parents remember learning, how they interpreted the information presented, how they implemented each skill, and their thought process around skill implementation. We also want parents to illustrate what the use of these skills looked like in their home and their understanding of how/why skills work. It will be important to go through the items slowly and methodically and help parents differentiate between different skills (e.g., use of rewards, bedtime fading).*

- A. Lead Question: You met with Lydia four times over the course of the study to learn skills to help your child sleep better. What do you remember learning as part of this treatment?
 1. Follow-Up Question 1: At the beginning of treatment you learned about normal sleep in children, what do you remember learning in this section?
 - a. Probe:
 1. Was any of the information presented a surprise for you?

2. Before hearing about the study on a scale of 1 (not at all important) to 10 (very important) how important did you believe that sleep was to your child's mental, emotional, and physical well-being?
3. Have your thoughts about the importance of sleep changed since you first heard about the study?
 - I. If so, how?
2. Follow-Up Question 2: In treatment you learned about sleep hygiene, techniques to help children sleep better by changing their environment and routines. What, if any, changes did you make in response to this section of treatment?
 - a. Probes:
 1. What is different about your child's bedtime routine since starting treatment?
 - I. Tell me about how you went about making these changes.
 - II. How did your child initially respond to these changes?
 - III. How are they currently responding to these changes?
 - IV. What is the same about your child's bedtime routine since starting treatment?
 2. What, if anything, is different about the time that your child goes to bed since treatment started?
 - I. Tell me about how you went about making these changes.
 - II. How did your child initially respond to these changes?
 - III. How are they currently responding to these changes?
 3. What, if anything, is different about the time that your child wakes up since treatment started?
 - I. Tell me about how you went about making these changes.
 - II. How did your child initially respond to these changes?
 - III. How are they currently responding to these changes?
3. Follow-Up Question 3: In treatment you learned how to reinforce positive sleep behaviors. What, if any, changes did you make in response to this area of treatment?

- a. Probes:
 1. What do you remember about the empathize and encourage statements? (e.g., “I know this is difficult for you but I know you can do it.”)
 2. How did you use these statements with your child?
 3. How did your child react to these statements?
 4. What do you remember about learning to use rewards?
 5. Did you use rewards to help your child improve sleep behaviors?
 - I. If so, tell me about how you went about implementing a reward system.
 - II. How did your child respond to rewards?
4. Follow-Up Question 4: In treatment you learned how to set limits, reduce accommodation, and help your child face his/her fears. What, if any, changes did you make in this area?
 - a. Probes:
 1. What limits did you set with your child (e.g., around electronics, napping, reassurance seeking, caffeine)?
 2. How did your child respond to these limits?
 3. What fears related to sleep did your child have (e.g., separation, dark, monsters)?
 4. How did you help your child deal with these fears?
 5. How did your child respond to facing his/her fears?

IV. **Treatment Effects:** *The intent of this section is to get feedback on the effects that parent training treatment had on the child’s sleep and what parents perceive as helpful and unhelpful. We are also interested in other ways that the treatment and changes in sleep have potentially impacted functioning more generally. When asking these questions make sure that it is clear that you are interested in the larger effects of treatment, not changes related to other things (e.g., school ended so now child is less stressed).*

- A. Lead Question: What do you think about the work that you and Lydia did building skills in treatment?
 1. Follow-Up Question 1: What do you think the effects of learning these skills was?
 - a. Probes:
 1. In what ways do you think treatment helped?
 2. In what ways do you think treatment didn’t help?

2. Follow-Up Question 2: How has your child's sleep changed since starting treatment?
 - a. Probes:
 1. How, if at all, has your child's anxiety around bedtime changed since treatment started?
 2. How, if at all, has the time it takes for your child to fall asleep at night changed since treatment started?
 3. How, if at all, has how often your child wakes up during the night changed since treatment started?
 4. How, if at all, has where your child sleeps at night changed since treatment started?
 5. How, if at all, has when your child wakes up changed since treatment started?
3. Follow-Up Question 3: You completed four 30-minute phone check-ins with Lydia, what did you think about these calls?
 - a. Probes:
 1. In what ways was these calls helpful?
 2. Unhelpful?
4. Follow-Up Question 4: How has the sleep treatment positively or negatively impacted non-sleep areas of your or your child's life?
 - a. Probes:
 1. How has it impacted things at school?
 2. At home?
 3. In her/his social life?
 4. His/her anxiety?
 5. General family functioning?
 6. Your levels of stress?
5. Follow-Up Question 5: Now that treatment has ended what are your thoughts about continuing to use the skills that you learned?
 - a. Probes:
 1. On a scale of 1 (not at all important) to 10 (very important) how important is it to you that you continue to use these skills with your child?
 - I. Why? (e.g., what makes it a 7 instead of a 6)
 2. On a scale of 1 (not at all confident) to 10 (completely confident) how confident are you that you will be able to continue using the skills you learned with Lydia?
 - I. Why? (e.g., what makes it a 7 instead of a 6)
 3. Which skills do you plan on continuing to use?

4. What do you think will help you to continue using these skills?
5. What may get in the way of continuing to use these skills?
6. Follow-Up Question 6: If you were to start treatment over again, what might you change?
 - a. Probes:
 1. Why?

V. **Treatment-Related Factors:** *The intent of this section is to get feedback on other factors related to treatment effects and the ways in which skills learned may transfer to other parenting behaviors, interactions with siblings, and other caregivers. Skip any section that is not applicable (e.g., if parent has only one child skip the section about how the skills have changed interactions with siblings). If running short on time in the interview this section can be shortened by skipping the probes but the follow-up questions at least should be asked if possible.*

1. Follow-Up Question 1: Sometimes parents find that the skills fit very naturally with their parenting style while others find them difficult at first to integrate into their normal style. For example, some families use rewards for completing chores so this skill is easy to use for sleep and other parents like to be very actively involved in bedtime so withdrawing some attention at bedtime can be difficult. How did the skills you learned in treatment fit with your own family culture and parenting style?
 - a. Probes:
 1. In what ways were skills consistent with your parenting style?
 2. In what ways were skills inconsistent with your parenting style?
 3. How did similarities or differences between your parenting style and the skills taught affect your actions?
2. Follow-Up Question 2: How has the treatment impacted other areas of your parenting behaviors?
 - a. Probes:
 1. In what areas have you noticed these changes?
 2. If applicable, have you noticed any changes in your parenting behavior around your child's siblings?

3. Have you used any of the skills learned in treatment with your other children?
 - I. If so, which skills, which contexts, and what was the result?
3. Follow-Up Question 3: If applicable, how much was the child's other caregiver involved in using skills learned in treatment?
 - a. Probes:
 1. How did you share information learned in treatment with the child's other caregiver?
 2. How often was the other caregiver involved in bedtime?
 3. How often did the other caregiver use the skills learned in treatment?
 4. What barriers were there to involving the other caregiver in using skills learned in treatment?
 5. What could have made the process of involving the other caregiver in using skills easier?
4. Follow-Up Question 4: Has learning about sleep and skills to improve sleep changed your behaviors around your own sleep?
 - a. Probes:
 1. If yes, how has it changed your behavior?
 2. If no, why not? (e.g., did not have sleep issues before, skills are not applicable to adults)
 3. Do you think that your child's sleep was related to your sleep in any way?
 4. If yes, when did you first notice this connection?
 5. What do you think about this connection?
5. Follow-Up Question 5: Were there any major changes in the family during the course of this study? (e.g., family moved, child started school, one parent was away on business, vacation)
 - a. Probes:
 1. If yes, what were they?
 2. How do you believe that these changes impacted you/your child's sleep and your use of skills?
6. Follow-Up Question 6: Do you have any suggestions of things that should be added to treatment to help parents improve their child's sleep?

VI. Wrap up

- A. Are there any other thoughts you have about your work with Lydia that you would like to share?

- B. Thank you so much for taking the time to speak with me today. We really appreciate hearing your thoughts!

Appendix E: Post-Treatment Qualitative Interview- Child

Goals & Purpose: The interviewers will conduct semi-structured qualitative interviews to elicit children's feedback about the assessment and treatment during the study as well as suggestions for improving both. Broad-based probes will focus on the clinical interview about anxiety and sleep, the computer task to measure reward learning, sleep diaries, actigraphy, questionnaires, treatment components, changes made due to treatment, importance of these changes, confidence in using skills learned in treatment, and suggestions for improving treatment for other children. Questions will be open-ended when possible to encourage informants to elaborate on issues they consider important.

Data Collection & Preliminary Analysis: Interviews will be conducted by study staff in a private treatment room at the Center for Anxiety and Related Disorders. Interviews will be completed by clinical doctoral students or highly trained undergraduate research assistants, average 30 minutes, and will be audio recorded or video recorded and transcribed.

Data analysis will consist of applied thematic and content analysis. Immediately after the data collection appointment (within 24 hours or before the next interview, whichever is sooner), the interviewer will review her handwritten notes taken during the interview and briefly summarize the interview dynamics and possible themes for further study. Once the interviews are transcribed, the person coding the transcripts will expand the summary based on content and thematic analysis. Participants will be identified by their study ID number in reports and transcripts. The interview guide and intent statements will serve as the basic structure for organizing the final report. Key points will be summarized on a question-by-question basis.

Equipment & Materials: The interviewer will take two digital recorders, extra batteries, and hard copy of the interview schedule (for note-taking). The interviewer will also schedule the video recording for the clinical room selected for this interview.

Intent Statements: Each section of this guide is preceded by an intent statement which explains the type of data that the section has been designed to gather. Intent statements are designed so that all interviewers focus their discussion on the same areas of interest and thus gather similar data. This will allow the interviewer to deviate from the script for a more naturally flowing conversation while also ensuring that relevant themes are being captured.

Interview Questions: There are two primary types of questions noted in the agenda: open-ended, "lead" questions and probes which tend to be less open-ended. Both lead questions and probes are intended to guide the interviewer to gather all the data investigators are seeking. They are not meant to be used verbatim. Interviewers should consider the best way to gather the data and rephrase questions to address intents and

local cultural norms and practices. Specifically, interviews should use different developmentally appropriate language if the child does not appear to understand a prompt.

Interview Schedule

- I. **Preliminary Information & Introductions:** *The intent of this portion of the agenda is to welcome the child and make him/her as comfortable as possible. Explain there are no right or wrong answers, no one will get mad at them for anything that they say, and that we will be taking notes but will not put their name on any of their answers. The child may need some time to warm up so make sure to spend some time at the beginning asking easier questions about things that they might find fun. If a child gets stuck on a question you can move on and briefly come back to the question at a later time. If a child is still confused by the question after you reword it or is not able to come up with an answer you may skip that question. If the child appears fatigued or distracted, make sure to ask if they would like to take a short break.*

A. Introductions & Purpose of Interview:

1. Thank you for meeting with me to talk about how your sleep has been and what you and your parent tried to make it better. My team is going to take what you tell me to get better at helping parents and kids build helpful sleeping habits. There are no right or wrong answers to any of these questions.
2. I'm going to ask you about the assessment, when you came into our Center the first time to answer questions about your sleep and how you are feeling. Next I will ask you about the things you and your parent tried to make your sleep better and if you have any other ideas on how to help kids like you sleep better.

B. Housekeeping:

1. You can take a break or ask to turn off the recorder at any time.
2. Please tell me if you feel uncomfortable for any reason.
3. You can stop the interview at any time – you do not have to keep going if you choose not to.
4. If I ask you something that doesn't really make sense, please tell me, and I can ask the question in another way.

C. Audio recordings:

1. Your thoughts and experiences are very important to me. Therefore, I would like to record our conversation. This is so that I remember what you say later on.

2. I am going to turn on the audio recorders. Try to just ignore the recorders.

D. **Warm up:** I'd like to start by getting to know you a little bit first.

1. How old are you?
2. What grade are you in school?
3. Who is in your family?
4. What kind of things do you like to do for fun?

II. **Assessment:** *The intent of this portion of the interview is obtain an understanding of the child's experience during the assessment. The purpose of gathering this information is to understand how to improve assessments and also get a sense of the child's understanding of why each assessment component was used. Examples of details to gather for each assessment component are initial thoughts about the assessment component, any concerns about the assessment, any way to improve aspects of the assessment or compliance with instructions, and ideas about why this assessment was conducted.*

A. Lead Question: When you first came to our Center and today you were asked to fill out a few questionnaires, talk about what has been going on with your sleep, and do a computer task. What did you think about doing those things?

1. Follow-Up Question 1: You did a computer task at the first assessment and again today where you looked at cartoon faces and decided which had a larger mouth. What did you think about this computer task?

a. Probes:

1. Did you feel frustrated, bored, happy, or any other emotion during the task?
2. Did it feel like it was too hard, too easy, or just right?

2. Follow-Up Question 3: During the study you wore a watch to track your sleep. What did you think about wearing this watch?

a. Probes:

1. What did you like about wearing the watch?
2. Was did you not like about wearing the watch (e.g., uncomfortable, fell off)?
3. Is there anything you think would make wearing the watch easier?

III. **Treatment Components:** *The intent of this section is to get feedback on the parent training treatment from the perspective of the child. We want to hear about what changes the child noticed and what they thought about these changes. We also want children to illustrate what the use of these skills looked like in their home and their understanding of why the changes were made. It will be important to go through the items slowly and methodically and help children differentiate between different changes.*

A. Lead Question: What do you think about your parent coming in to the center to learn how to help you sleep better?

1. Follow-Up Question 1: Why do you think sometimes kids get help to sleep better?

a. Probes:

1. What kinds of things might get better or worse after a kid is sleeping better?

2. Follow-Up Question 2: What kinds of things did you and your parent try to help you sleep better?

a. Probes:

1. What things that you and your parent tried were helpful?

2. What things weren't as helpful?

3. Follow-Up Question 3: What changes have there been in how you get ready for bed recently?

**After the participant has listed the changes, for each change listed go through each probe using their own language (e.g., "Mommy reads me a book and then goes to bed in her room instead of staying in my bed until I fall asleep.") If they cannot think of any changes ask if there have been changes in the order they do things, if they do them with or without parents now, if anything takes a shorter or longer time.*

a. Probes:

1. What has it been like to make _____ change?

2. Are there things that you like or don't like about it?

3. What did you first think when your parent said you were going to make __ change?

4. What do you think of this change now?

5. Why do you think that you and your parent made this change?

4. Follow-Up Question 4: What changes have there been in where you sleep recently?

**** After the participant has listed the changes, for each change listed go through each probe using their own language (e.g., “Sometimes I sleep by myself in my bed and sometimes I sleep in my bed first and then go into my parent’s bed.”)***

a. Probes:

1. What has it been like to make _____ change?
 2. Are there things that you like or don’t like about it?
 3. What did you first think when your parent said you were going to make __ change?
 4. What do you think of this change now?
 5. Why do you think that you and your parent made this change?
5. Follow-Up Question 5: What changes have there been in what you do if you wake up in the middle of the night recently?

**** After the participant has listed the changes, for each change listed go through each probe using their own language (e.g., “When I wake up I try to go back to sleep by myself but sometimes I get scared and go to my mom.”)***

a. Probes:

1. What has it been like to make _____ change?
 2. Are there things that you like or don’t like about it?
 3. What did you first think when your parent said you were going to make __ change?
 4. What do you think of this change now?
 5. Why do you think that you and your parent made this change?
6. Follow-Up Question 6: What changes have there been in what you do when you wake up in the morning recently?

**** After the participant has listed the changes, for each change listed go through each probe using their own language (e.g., “I want to lay in bed but my dad makes me get up and dressed right away and then I can watch 10 minutes of TV.”)***

a. Probes:

1. What has it been like to make _____ change?
2. Are there things that you like or don’t like about it?
3. What did you first think when your parent said you were going to make __ change?
4. What do you think of this change now?

5. Why do you think that you and your parent made this change?

IV. **Treatment Effects:** *The intent of this section is to get feedback on the effects that parent training treatment had from the child's perspective. We are also interested in other ways that the treatment and changes in sleep have potentially impacted functioning more generally. When asking these questions make sure that it is clear that you are interested in the larger effects of treatment, not changes related to other things (e.g., school ended so now child is less stressed).*

- A. Lead Question: How do you think your sleep has been in the last two weeks?
 1. Follow-Up Question 1: Do you think that your sleep has gotten better, stayed the same, or gotten worse in the last month?
 - a. Probes:
 1. Why do you think that your sleep has gotten better or worse?
 2. Do you think that your sleep will keep getting better, stay the same, or get worse now that your parent has learned some ways to help you sleep?
 - I. Why?
 2. Follow-Up Question 2: For some people, how well rested or tired they feel can change how easy it is to pay attention, what kind of activities they want to do, or how they feel overall. How do you think your sleep is related to other things in your life like school, home, and your friends?
 - a. Probes:
 1. Do you think that changes in your sleep made:
 - I. School better or worse in the last month?
 - II. Things at home better or worse in the last month?
 - III. Things with your friends better or worse in the last month?
 3. Follow-Up Question 3: Are there any other things that you think could help kids to sleep better?

V. **Wrap up**

- A. Are there any other thoughts about sleep or the things you tried with your parents to sleep better that you would like to share?
- B. Thank you so much for taking the time to speak with me today. We really appreciate hearing your thoughts!

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Curriculum Vitae

